EFDC_Explorer8.4 and EFDC+ Guidance
New Features and Functionality



DRAFT

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1 New Features of EE8.4 and EFDC+

Dynamic Solutions – International, LLC continues to develop new functions and capabilities of the EFDC_Explorer Modeling System. This latest release of EEMMS includes enhancements focused on habitat analysis tools, and time governed hydraulic structures. Several bug fixes and tweaks have been addressed. Major new features of EE8.4 and EFDC+ include the following:

- **Habitat analysis with critical limits time series**
- **Habitat Suitability Criteria (HSC) analysis with Instream Flow Incremental Method (IFIM)**
- **Time variable and rule-based control of sluice gates**
- **Rule-based control of the withdrawal/return boundary**

Minor Modifications and Bug Fixes

- Improved structure for EFDC+ output files for increased reading speed and logic.
- Spatially variable AHD and AHO
- Momentum option for the flow boundary type
- Groundwater linked to mass balance
- Elevation specified – Free tangential open boundary condition
- Whole channel rating curve option
- Improved calibration tools for statistics, plotting formats, sub-periods
- Ability to edit and save KML files in EE.
- Saved plots features

2 Habitat Analysis Tools

Habitat models are designed for a wide variety of planning applications where habitat information is an important consideration in the decision-making process. EEMS now supports two new types of habitat analysis tool. These are in addition to the existing temperature limit and depth criteria tool in *ViewPlan* that was initially implemented to support studies on manatee populations in the Chassahowitzka Estuary. The two new tools generate time series critical limits, as well as support the state-of-the-art Instream Flow Incremental Method. These two methods are described below.

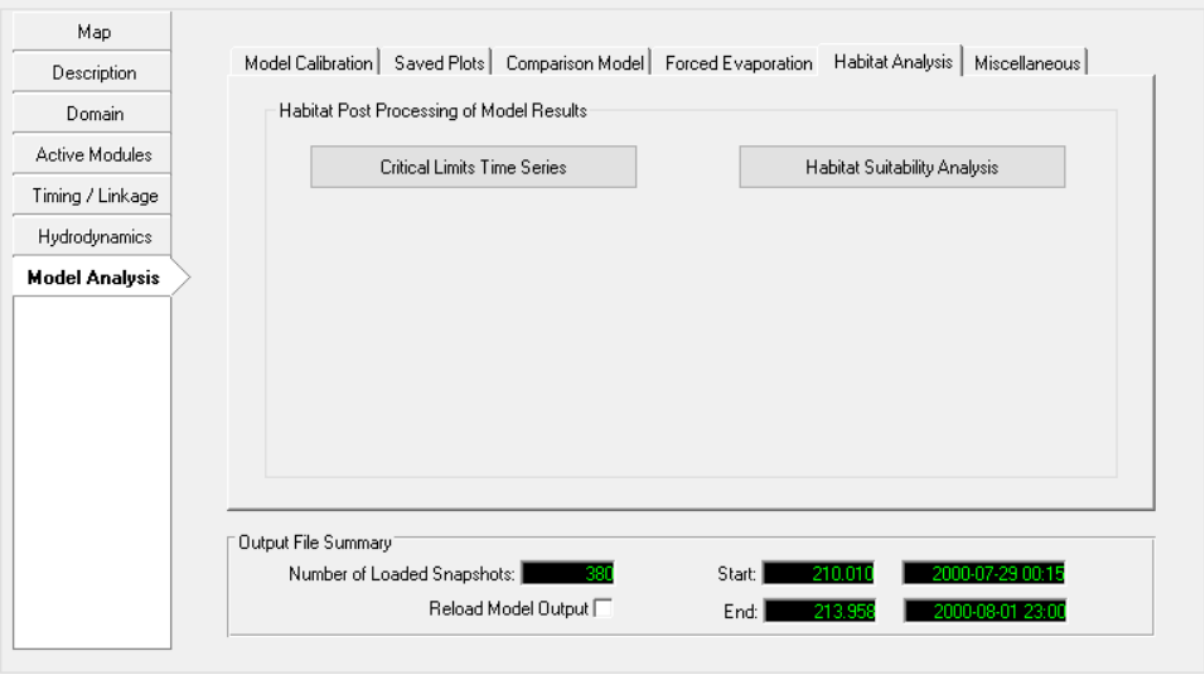


Figure 1. Model Analysis: Habitat Analysis form.

2.1 Critical Limits Time Series

The *Critical Limits Series* form is shown in Figure 1. The user may define up to five different parameters for the critical limits. These can be selected from any parameter that is currently activated in EE. For each dropdown list under *Parameter* the user selects the parameter of interest and then defines the minimum and maximum values for the critical limits. In the layer box a range of options are provided: single layer (enter layer number), a range of layers (eg 1-2), average of a series of layer (eg 2:6), or depth averaged (0). Once criteria for critical limits have been entered it is recommended the user save the settings. Selecting the *Save* button, the user is prompted to input the file name. EE will create a folder called #habitat to store all related files, so by default the new critical limits will be saved into this folder. Configuration files for critical limits series use ".HCL" as the extension.

The *Criteria ID* will be used to manage the different species. The *Criteria ID* is appended to the file name "Hab_Results" of the data file extracted when generating an initial time series, as described further below.

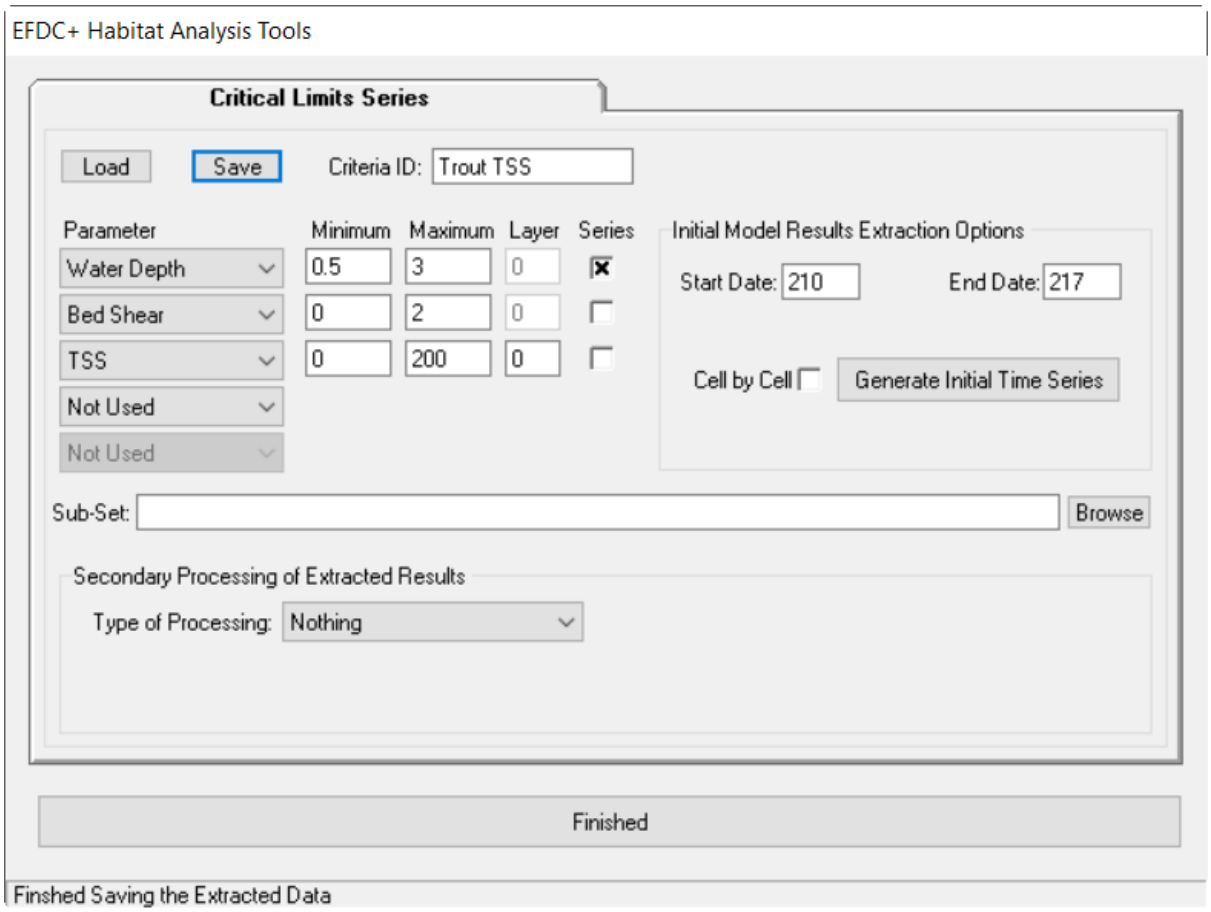


Figure 2. Habitat Analysis: Critical Limits Series form.

The critical limits options act as a series of toggles for each cell in each layer of the model domain. Each cell will meet either some of the criteria, or all of criteria defined. If a cell doesn't meet the all the criteria it is set to zero, whereas if a cell does meet all the criteria it is set to one. If a cell is dry it is also set to zero. Using the *Series* checkbox, the user turns on the generation of output for each parameter. Only one parameter may be output at a time.

The user should select the start and end date for the time series extraction in the *Initial Model Results Extraction Options* frame. Once this has been set the user can select the

Generate Initial Time Series button. EE will then tell the user how many cells are in the domain and check if the user wants to proceed with calculating the time series. Selecting the *Cell by Cell* check-box will mean that every cell is calculated for every time step. Note that it is also possible to generate the time series based on only a sub-set of the model domain by browsing to and selecting a polyline file with the *Sub-Set* box.

EE will extract all the cell information for each parameter and save it in a post processing file in the #habitat folder based on the criteria ID. In this example the file will be called Hab_Results_Trout01.dat. This file records the composite time, salinity, net weighted criteria (weighted average 0 and 1 of all the cells that meet the bathymetry criteria), area, and volume.

An example of the time series that EE displays is shown in Figure 3.

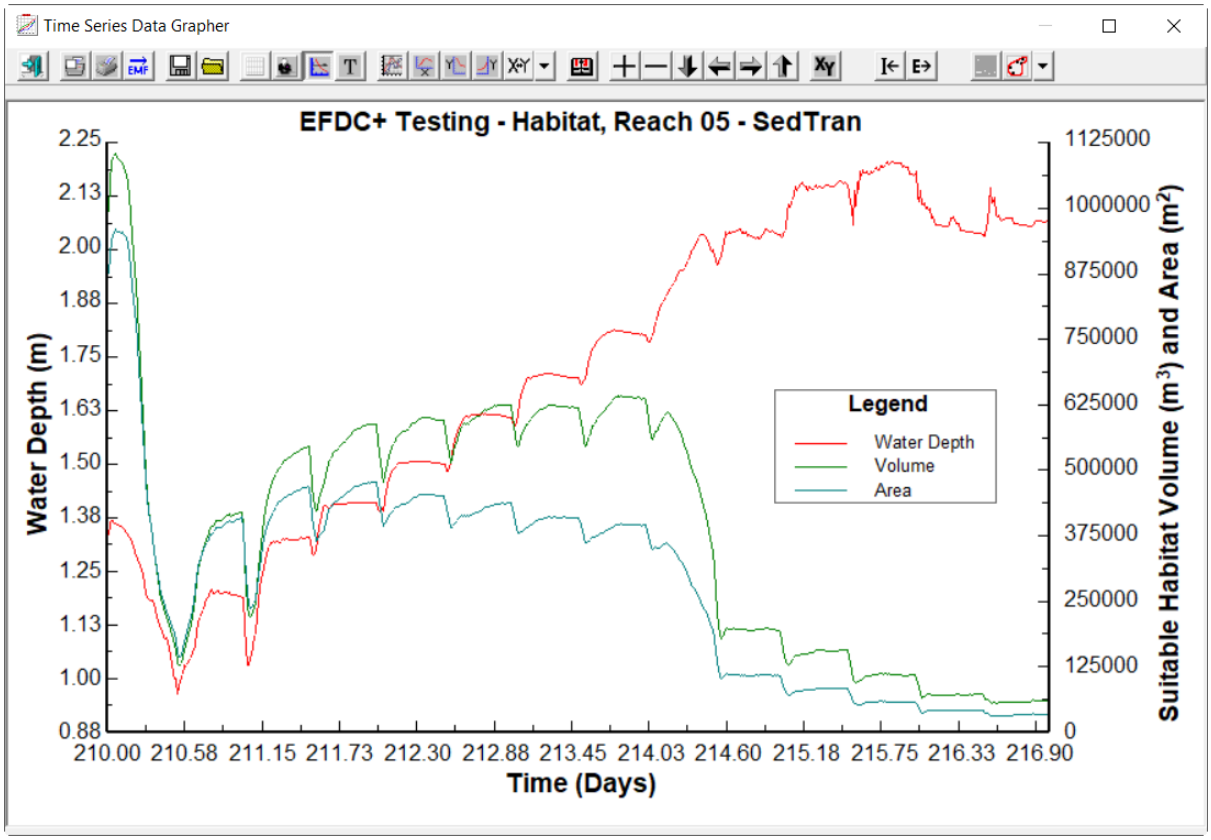


Figure 3. Habitat Analysis: Critical Limits Series time series.

The results in the time series plots present a composite time series of velocity and the volume and area that met these criteria. The user should export the time series as it is not automatically saved by EE.

Once a time series has been generated, and the post-processing files have been extracted, a new frame is displayed, *Secondary Processing of Extracted Results.* This option provides the user with further post-processing options. The drop-down menu has the following options: *Domain Average, Running Averages*, and *Single Cell Time Series*. For running averages, the user must specify the number of days to be applied for calculation of the running average.

2.2 Habitat Suitability Criteria Analysis

The Habitat Suitability Criteria (HSC) models provide an objective, quantifiable method of assessing the existing habitat conditions for a given aquatic species within a study area by measuring how well each habitat variable meets the habitat requirements of the species by life stage. This serves as an objective basis for predicting probable project impacts, documenting post-project impacts, and guiding habitat protection, mitigation, enhancement, and management decision.

Instream Flow Incremental Method (IFIM) is a method of determining the HSC, and is one of the most widely used instruments in the world for assessing the effect of flow manipulation on river habitats. Originally developed by the US Fish and Wildlife Service (Bovee, 1982), USA EPA describes IFIM as integrating "concepts of water-supply planning, analytical hydraulic engineering models, and empirically derived habitat-versus-flow functions to address water-use and instream-flow issues and questions concerning life-stage-specific effects on selected species and the general well-being of aquatic biological populations. The method has undergone continual refinement by the Fort Collins Science Center scientists and has remained a state-of-the-art tool. A major component of IFIM is a collection of computer models called the Physical Habitat Simulation Model (PHABSIM). This model incorporates hydrology, stream morphology, and microhabitat preferences to determine relations between streamflow and habitat availability" (EPA, 2018).

At this stage, however, the PHABSIM models to not simulate water quality. PHABSIM concentrates on simulation of physical habitat based on depth, velocity, and a channel index. In their manual on PHABSIM, Milhous and Waddle (2012) refer users to other sets of models, such as the QUAL-2 model for approaches to simulate water quality constituents. Furthermore, PHABSIM, River2D and similar IFIM models are restricted to 2D simulation. With EEMS now enhanced to perform the IFIM analysis, it serves as a fully highly advanced 3D PHABSIM, incorporating water quality, sediment, toxics and many other highly relevant sub-models.

Habitat availability with IFIM is measured using an index called the weighted useable area (WUA). This is the wetted area of a stream weighted by its suitability for use by an organism. In EE WUA is calculated using the product of the habitat suitability indices, which are further described below:

$$WUA = \sum_L A_L C_L \prod_1^n s_i$$

Where

- A_L is the area of the cell
- C_L is the habitat suitability index for the channel index associated with the cell substrate and/or vegetation, if not used or needed a value of 1.0 is used,
- L is the cell index
- S_i is the habitat suitability index for parameter i for up to n parameters.

Another measure is the weighted usable volume, defined as:

$$WUV = \sum_L V_L C_L \prod_1^n s_i$$

Where

- V_L is the volume of the cell

The interface for the IFIM approach to configuring habitat suitability criteria in EE is shown in Figure 4. When saving and loading the configuration files for habitat suitability criteria EE uses ".HSC" as the extension.

The *Criteria ID* will be used to manage the different species. The *Criteria ID* is appended to the file name "HSC_Results" of the data file extracted when generating an initial time series. For the example shown in Figure 4 the extracted file will be called "HSC_Results_Walleye – Open Water.dat"

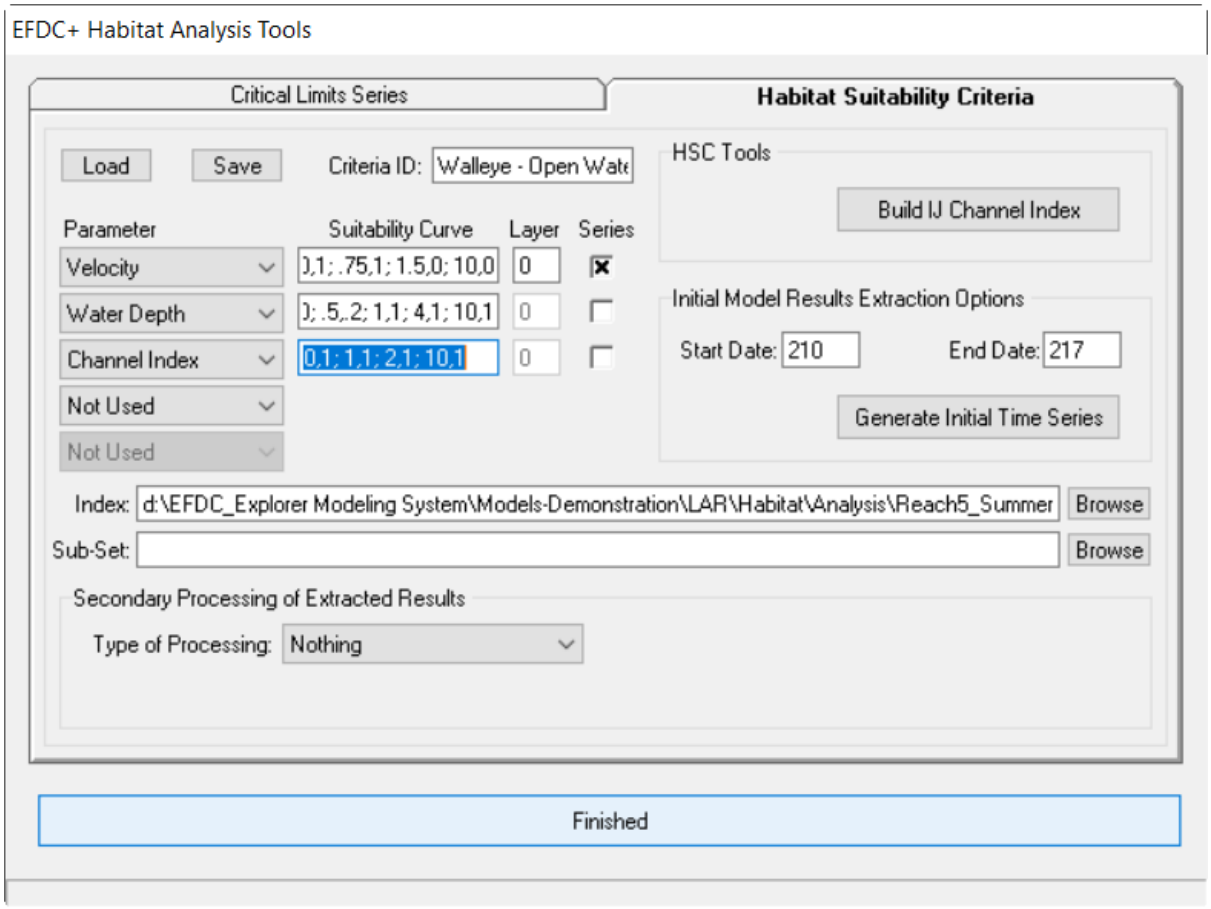


Figure 4. Model Analysis: Habitat Suitability Criteria form.

Parameters that may be defined include bottom elevation, water depth, bed shear, channel index, velocity, fraction of light, cohesives, and TSS. The general approach for calculating the weighted usable area (WUA) for habitat is quite similar to that for the critical limits approach. One significant difference in the model setup is that the inflows are incremental, or stepped in IFIM approach. There are two main differences in the post-processing of the data: defining suitability index curves, and specifying a channel index. These are described below.

2.2.1. Suitability Index Curves

Rather than using simple maximum and minimum limits for a parameter, as was the case for critical limits, the HSI approach uses Suitability Index (SI) curves. These SI curves are constructed by quantifying field and laboratory information on the effect of each habitat variable, such as temperature, dissolved oxygen, spawning gravel size, and siltation, on the growth, survival, or biomass of a species by life stage. The SI curves assume that increments of growth, survival, or biomass plotted on the y-axis can be directly converted into an index of suitability on the x-axis. The index of suitability ranges from 0.0 to 1.0 with 0.0 indicating unsuitable conditions and 1.0 optimal conditions. Measurements of each habitat variable taken at the proper time in the field can be applied to the SI curves to assess the suitability of the variable in meeting the habitat requirements of the species by life stage

To facilitate the input of these curves, EE provides a tool to create and view SI curves. After selecting a parameter from the dropdown menu, selecting the *Suitability Curves* text box and pressing *Enter* will open the *XY Data Editing Tool*, which is a spreadsheet type form with a corresponding graph as shown in Figure 5. The user may choose to initially develop the curves in a spreadsheet and copy them into the *Suitability Curve* form where they will be stored and plotted. Alternatively, they can be typed directly into the form as (parameter, suitability). These x, y values are comma delimited, with each pair then separated by a semi-colon.

After defining the SI curves the user should select the layer range as described in the *Critical Limits* section. It is required to select the checkbox to assign which series is to be plotted, and the start and end date of extraction from the model, before selecting *Generate Initial Time Series.*

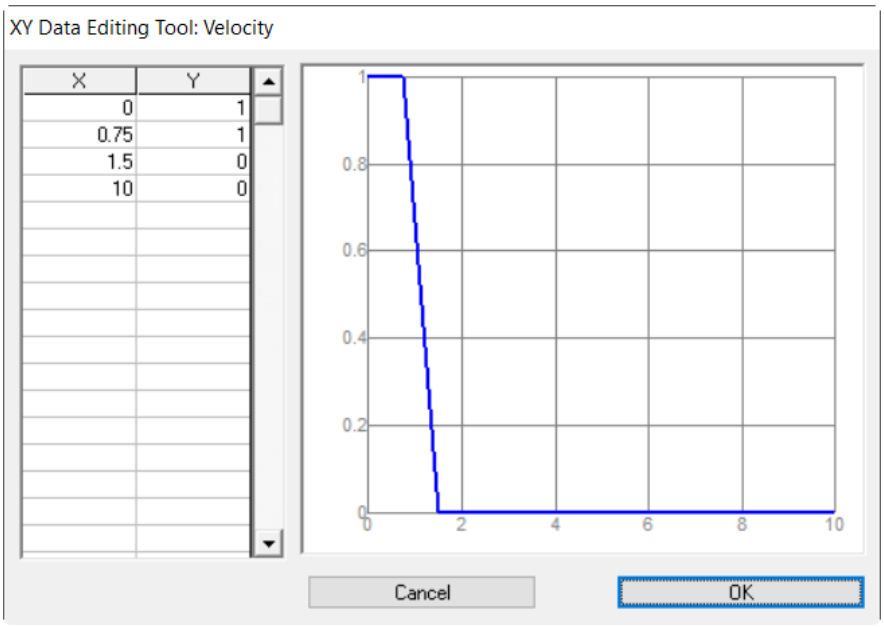


Figure 5. Editing tool for suitability index curves.

2.2.2. Channel Index

Another option available in the HSC approach, which is not in the *Critical Limits* option, is the defining of a channel index. The purpose of the channel index is to allow the user to evaluate the habitat suitability based on nature of the channel bed, which may vary in different regions of the stream, for example with sand, silt or gravel. This index can be set based on different locations in the channel that the species may be occupying. The channel index is a numeric value that has been assigned to each cell. EE allows the user to assign any value to a cell, though typically they range from 0 to 1, or 0 to 100. *Channel index* is one of the parameter options, and should be selected from the dropdown menu if required.

If the user doesn't have a channel index file already, they should first build the channel index with the *Build IJ Index* button. Selecting this will open the tool shown in Figure 6. In this form the user should browse to a channel index file from another tool such as River2D (.CHI) and import this file. River2D format is: "X, Y, Elevation, Substrate Code", where X and Y should be in UTM. However, EE will prompt the user for the column containing the channel index so other formats can also be imported. Alternatively, the user can generate this externally in a spreadsheet. EE will interpolate the channel index to the I, J grid. The user selects the name of the output file, which will be saved in the EE file format with extension .CI and of the type: I, J, Channel Index.

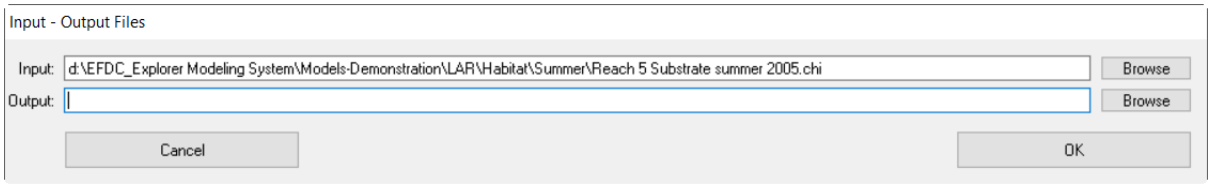


Figure 6. Habitat Suitability Criteria: Channel index tool.

Once the user has a channel index file already created they should browse to it in the *Index* box. As described in the critical limits section, it is also possible to generate the time series based on only a sub-set of the model domain by browsing to and selecting a polyline file with the *Sub-Set* box. If no file is selected in the *Index* option EE will effectively ignore this option by assuming the index is 1.0 everywhere.

The user should select the start and end date for the time series extraction in the *Initial Model Results Extraction Options* frame. Once this has been set the user can select the *Generate Initial Time Series* button. EE will then confirm the number of cells in the model domain and check if the user wants to proceed with calculating the time series.

To create time series EE will extract all of the cell information and save it in a post processing file in the #habitat folder with a name based on the *Criteria ID*. If the criteria ID is "Salmon01" then the file will be called HSC_Results_Salmon01.dat. This file records the composite time, salinity, net weighted criteria (weighted average 0 and 1 of all the cells that meet the bathymetry criteria), area, and volume.

Once a time series has been generated, and the post-processing files have been extracted, a new frame is displayed, *Secondary Processing of Extracted Results as shown in* Figure 8.

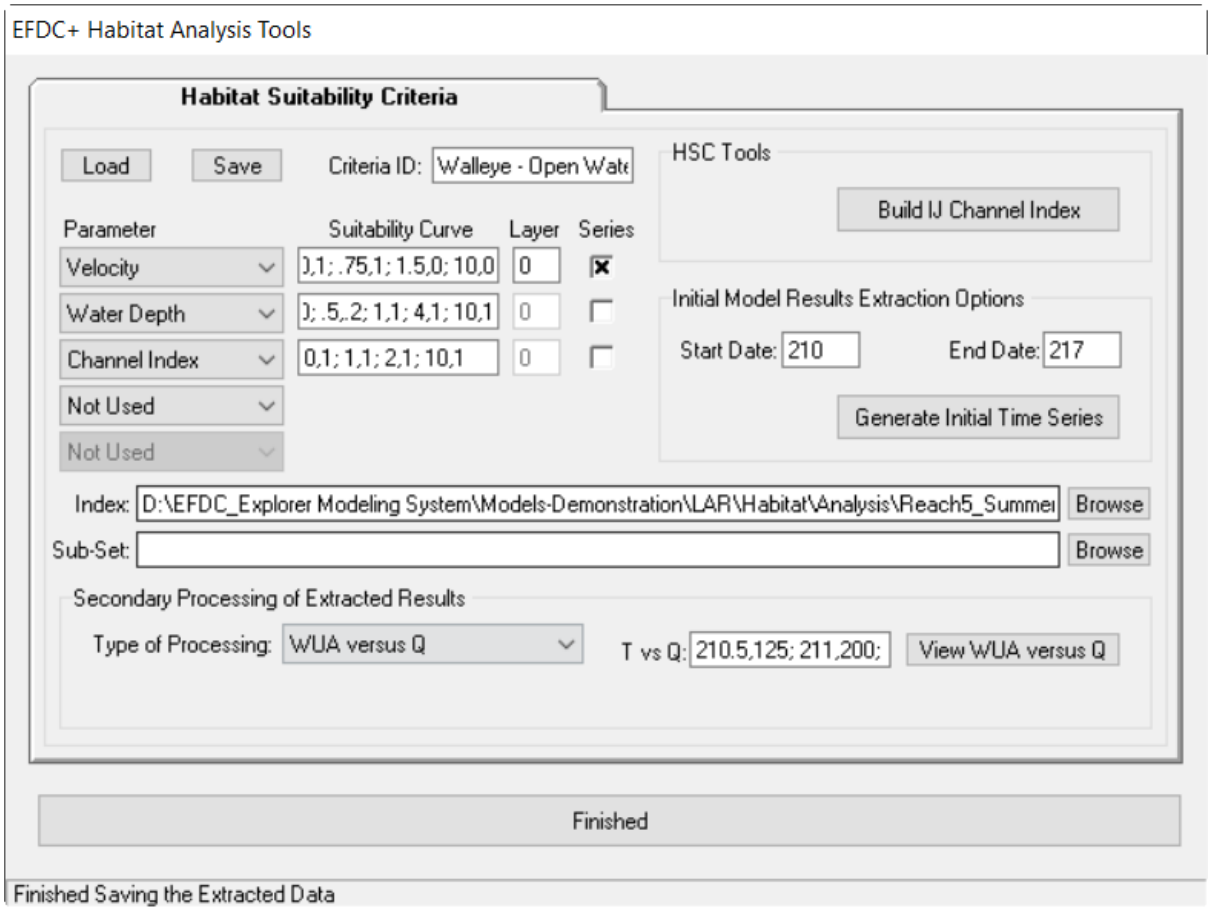


Figure 7. Suitability Criteria: Secondary processing.

The drop-down menu for *Type of Processing* has the following options:

- *WUA Time Series:* This is a time series of the weighted usable area
- *WUA vs Q:* For this option the user must create a time series versus flow curve. This curve should match the initial boundary conditions that are being applied to the model. The user should enter the time versus flow (T vs Q) as x, y points in the table. EE will extract the WUA for the flow at this time and plot WUA vs flow.
- *HSC Time Series:* This is a number from 0 to 1 that is weighted over the entire domain.
- *WUV Time Series:* time series of weighted usable volume
- *Volume Time Series:* time series of volume
- *Parameter Time Series:* Depending on which option with a check box has been selected (only one at a time may be selected), EE will extract the time series for that parameter

After selecting one of these options EE will display a button to the right of the menu. Selecting this button will display the time series such as that for the WUA time series shown in Figure 8. If the user has extracted the results previously in this session, and then wants to view another secondary processing type, EE will prompt whether to use the data in memory, or whether to extract the data again.

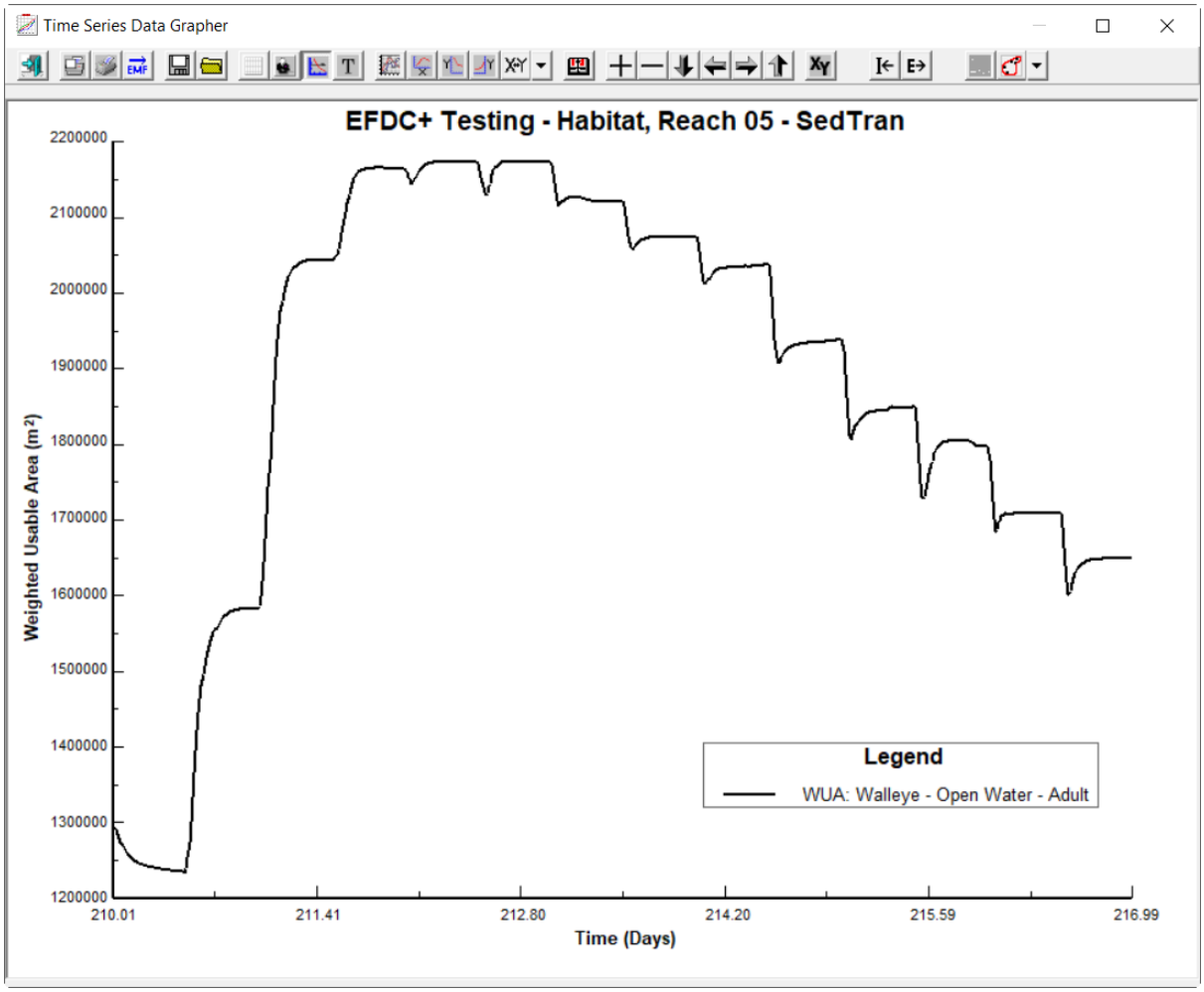


Figure 8. Weighted Usable Area time series.

2.3 Habitat Results in ViewPlan

In addition to the temperature limit and depth criteria tool in *ViewPlan,* initially implemented to support studies on manatee populations in the Chassahowitzka Estuary, the critical time limits and habitat suitability criteria analysis are also available to be displayed in 2D plan view. When habitat results are loaded into EE the user may select Alt-H in *ViewPlan* to select one of the habitat tools as shown in Figure 9. Select 1, 2, or 3, for the *Thermal and Depth Criteria Limits, Critical Limits,* or *HSC* options respectively.

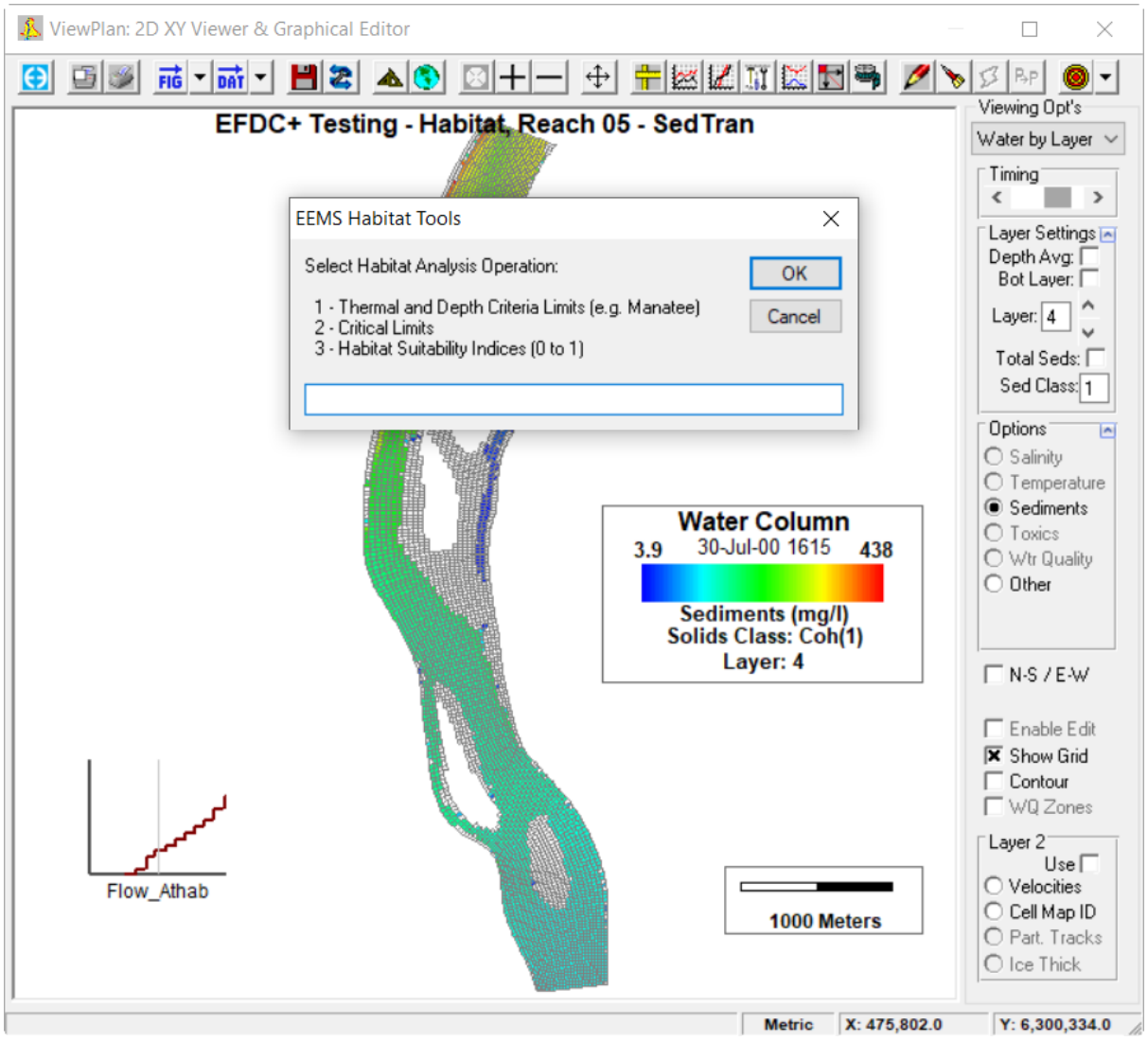


Figure 9. ViewPlan: Habitat tools.

Selecting option 2 or 3 will cause EE to prompt the user for the habitat limits or criteria file. EE will then display areas satisfied by the criteria, as shown in Figure 10. This option is available for all the parameters that may be set in the habitat critical limits, or suitability criteria. If a cell is dry it is shown as grey. For the critical limits the default color ramp is white (0) to green (1). For habitat suitability criteria the same principle applies, however shades of green from 0 to 1 are displayed.

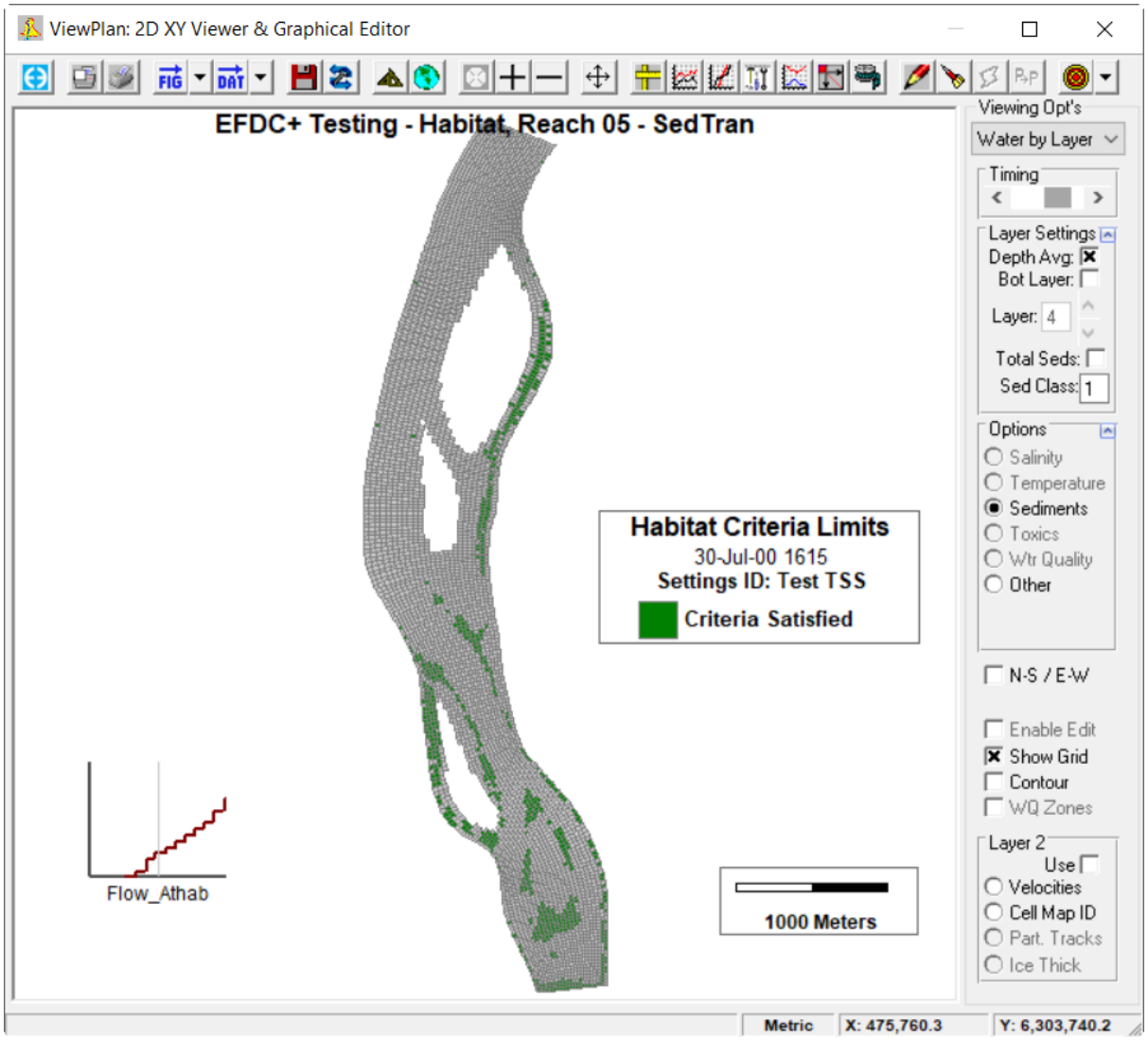
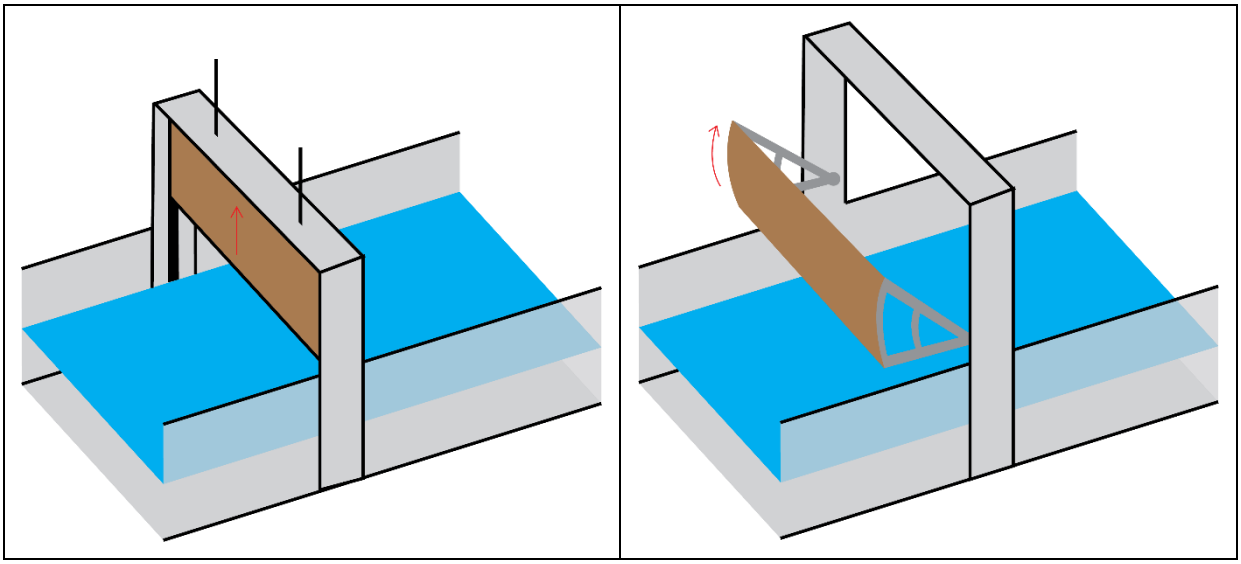


Figure 10. ViewPlan: Habitat criteria limits.

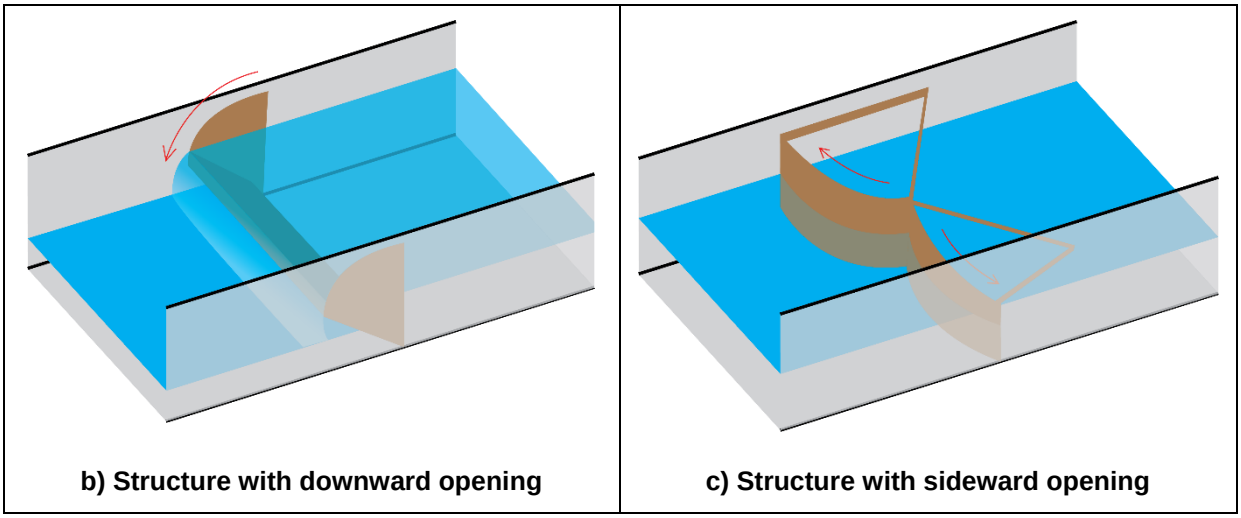
3 Time and Rule-based Control of Sluice Gate

The hydraulic structures boundaries have now been enhanced to simulate the opening and closing of sluice gates and similar hydraulic structures. Currently two types of sluice gate operation are supported: operational time series, and operational rules.

With the operational time series option, the gate is opened or closed during the simulation based on a defined time series of the operational state and settings such as rate of gate opening, and maximum opening height. Examples of the types of gates may be shown in Figure 11.



a) Structures with upward opening



b) Structure with downward opening

c) Structure with sideward opening

Figure 11. Operation for different types of gates.

To setup these options the user should open the *Hydraulic Control Structure BC* form shown in Figure 12. This is accessed from the *Domain* | *Boundary Conditions* | *Edit/Review* button and then selecting a hydraulic structure boundary (type 2).

In the *Flow Derived From* dropdown menu a new *Upstream Elevation Whole Channel Rating Curve* option has been added which is discussed in Section 5.4. The options available in previous versions of EE that use rating curves are still set in the same way in the *Cell Options* | *Rating Table.* Also listed here are the equation based hydraulic structure options:

- Upstream Elevation Whole Channel Rating (new feature described in Section 5.4)
- Upstream Depth
- Elevation Difference
- Elevation Difference with Flow Accelerations
- US and DS Elevations
- Upstream Depth with Low Chord
- Elevation Difference with Low Chord
- Equation: Culvert
- Equation: Sluice Gate
- Equation: Weir
- Equation: Orifice

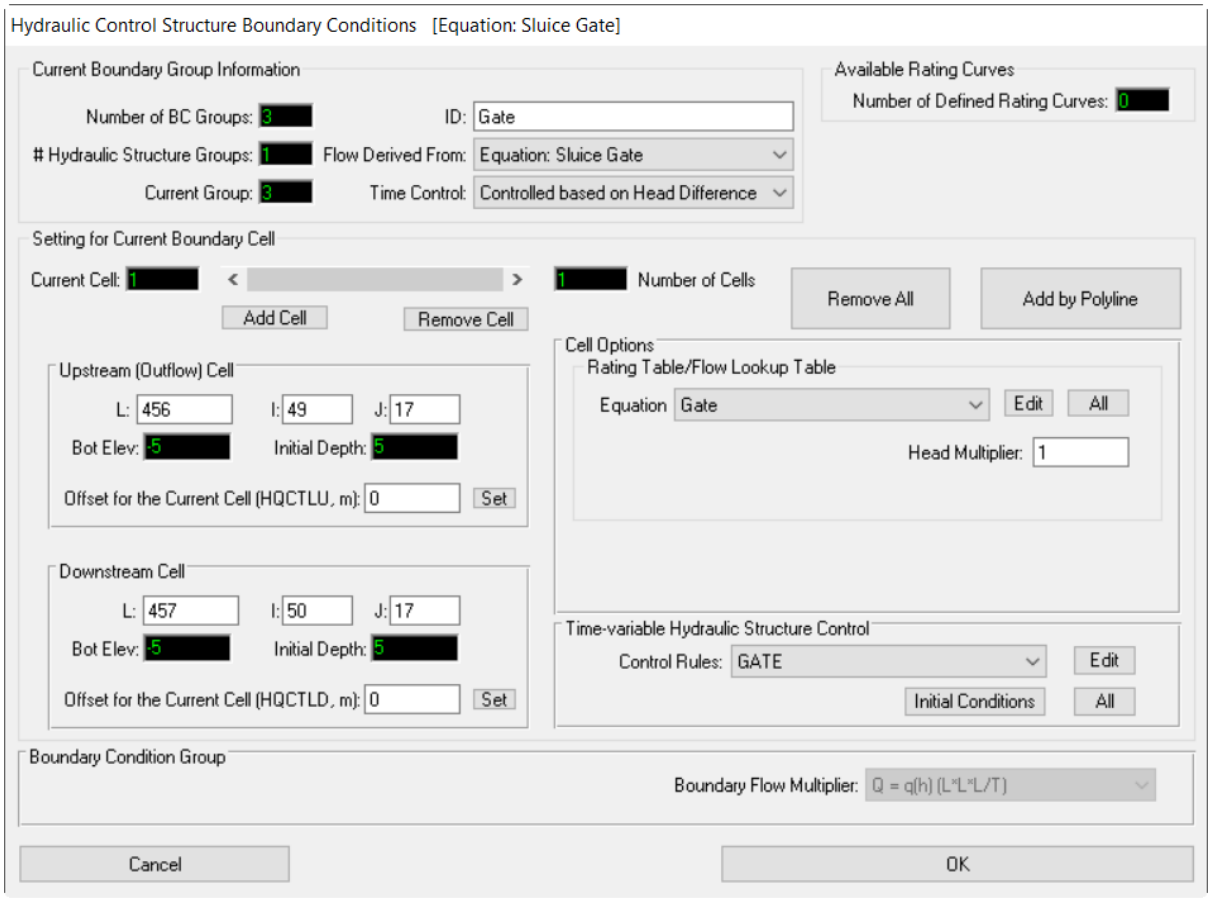


Figure 12. Hydraulic Structures: Sluice gate controls.

When using the equation based hydraulic structures, if the user selects *Equation: Sluice Gate*, a new *Time Control* dropdown menu will be displayed as shown in Figure 12. The options available in this menu are as follows:

Name	Meaning
Uncontrolled Structure	The hydraulic structure is uncontrollable type. The old method of flow computational using lookup table or equation will be used
Controlled using Time-Series	The operation of the hydraulic structure is controlled using a time-series which defines the changes with time of gate openings or rating curves
Controlled based on Upstream Elevation	The operation of the hydraulic structure is controlled using control rules defined based on water surface elevation at an upstream location. This location can be different from the upstream cell of the structure.
Controlled based on Head Difference	The operation of the hydraulic structure is controlled using control rules defined based on the difference of water surface elevations at two locations upstream and downstream of the structure. These locations can be different from the upstream and downstream cells of the structure.

These three new options are described below.

3.1.1. Time-Series Control

When the flow control type of *Controlled using Time-Series* is selected for sluice gate, the user can select which time-series will be used for the structure by selecting the corresponding item in the *Table* dropdown list in *Time Variable Hydraulic Structure Control* group (Figure 13). To create new time series, or edit an existing one, the user can click on *Edit* button beside the *Table* dropdown list to edit the control time-series which is shown in Figure 14. Selecting the *Show Params* button provides the user with more control over which parameters from the time series will be used to control the gate. Parameters that can be selected for the sluice gate include the initial opening height, opening width, sill level change, number of units, flow discharge and rating curve. A setting of “1” means that this parameter is used in control calculations. A setting of “0” means this parameter is ignored. In this example only gate height is set to 1. Other parameters are left to 0, meaning only the opening height of the gate will change with time.

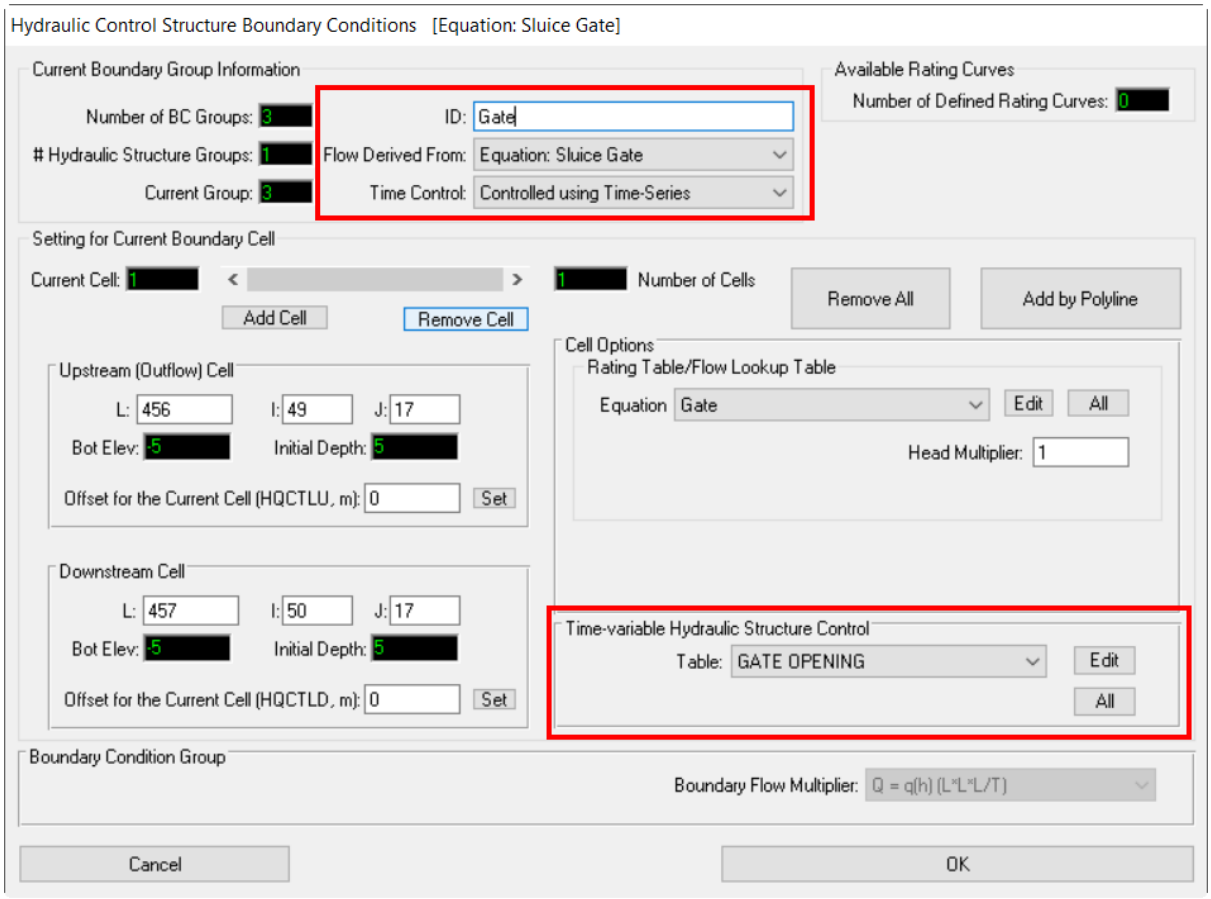


Figure 13. GUI for hydraulic structure control using a time-series.

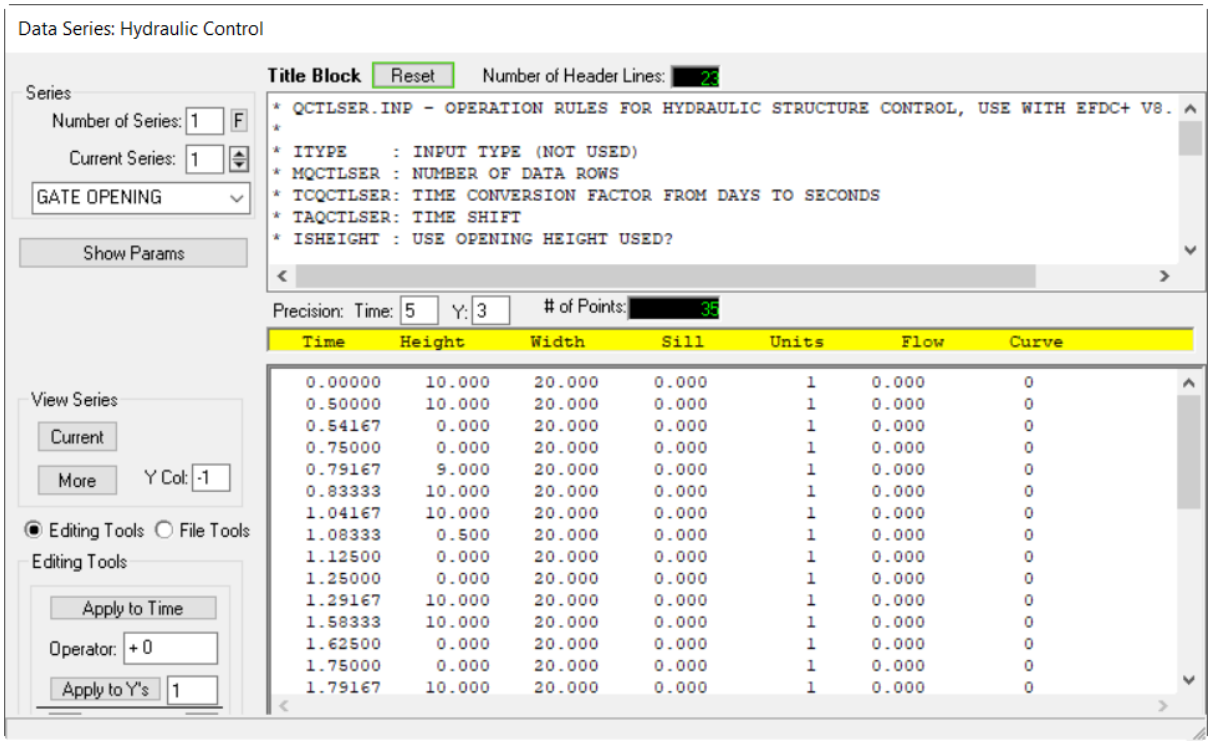


Figure 14: Data Series: Time series for sluice gate control.

3.1.2. Control Rules from Upstream Elevation or Head Difference

When the flow control type of "*Controlled based on Upstream Level*" or "*Controlled based on Head Difference*" is selected, the user can select which control rules will be used for the structure by selecting the corresponding item in the *Control Rules* dropdown list in *Time Variable Hydraulic Structure Control* group shown in Figure 15.

To create a new set of control rules, or edit an existing set, the user can click on *Edit* button beside the *Control Rules* dropdown.

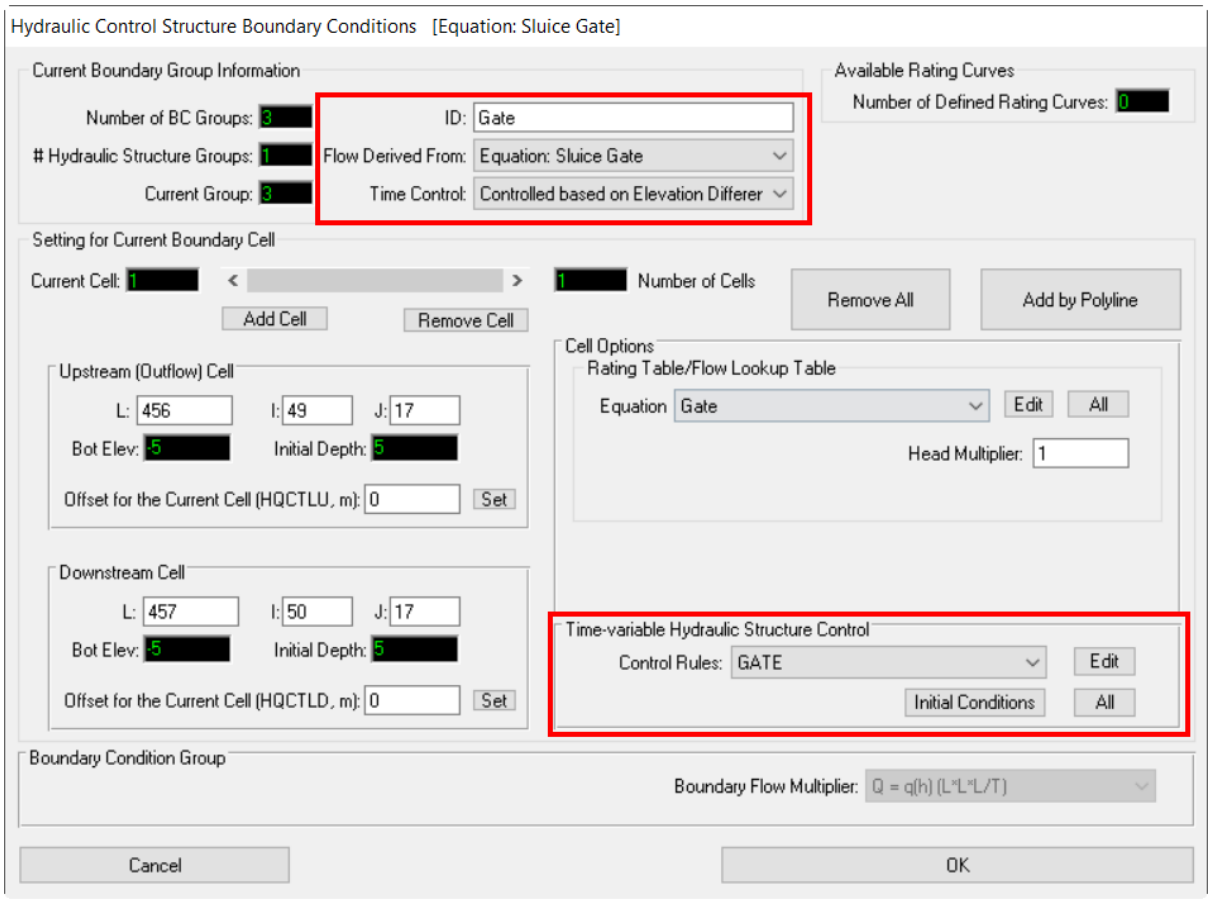


Figure 15. GUI for hydraulic structure control using operational rules.

The hydraulic structure control rules are shown in Figure 16. Here the user can configure the number of sets of rules with *Number of Rules*, and then scroll between the sets with *Current Rule* menu. For each set of control rules, the user must configure which parameters the rules are to be set for: opening height, opening width, and sill level (height above bottom of structure) change. The actual rules require the following settings:

- Control value: the elevation or (head difference) which will trigger the gate to start to open or close
- State: the direction of the opening or closing. Opening is configured with a "1" and closing is configured with "0"
- Height: opening height to which the structure can rise or fall
- Width: the opening width (m) of the gate at (for sideward opening).
- Sill Level Change: the sill level change (m) of the gate. Default is zero.
- Rate: the rate of increase or decrease of the gate opening.

EE uses the "state" of the structure to allow the user the high level of control required to simulate realistic scenarios. The logic of the process is that if the state of the structure is "1" then it is currently opening, and the rate of opening determined by the *Rate.* This will occur

from that trigger level until it reaches another trigger level, or it reaches the *Height* assigned (presumably the maximum opening). Once it reaches the maximum opening the rate no longer applies and the final flow rate is maintained. When a trigger is reached with state of "0" the gate will close, and flow decrease at the set *Rate.* This continues until it reaches the *Height* assigned (presumably fully closed).

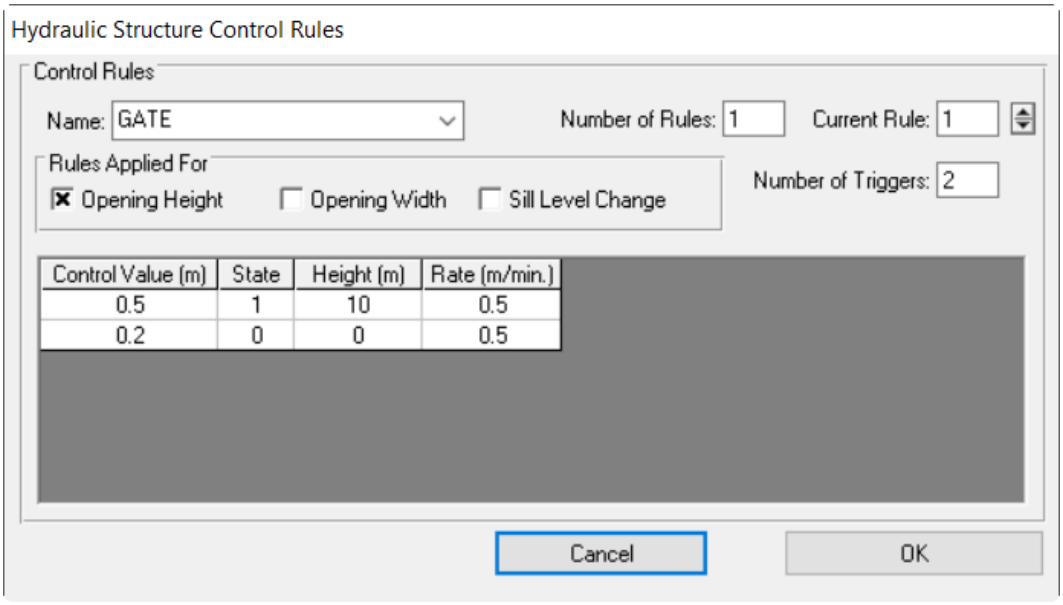


Figure 16. Operational rules configuration for sluice gate.

Figure 17 shows the form for setting the initial condition of the hydraulic structure. As described above, the location of the hydraulic structure trigger does not have to be in the same cell as the actual structure. Here the user can configure the upstream and downstream locations of the control triggers in the *Time Control Locations* frame. The user should also configure the initial conditions: whether the gate is opening or closing, and the initial height width and sill level change, if any.

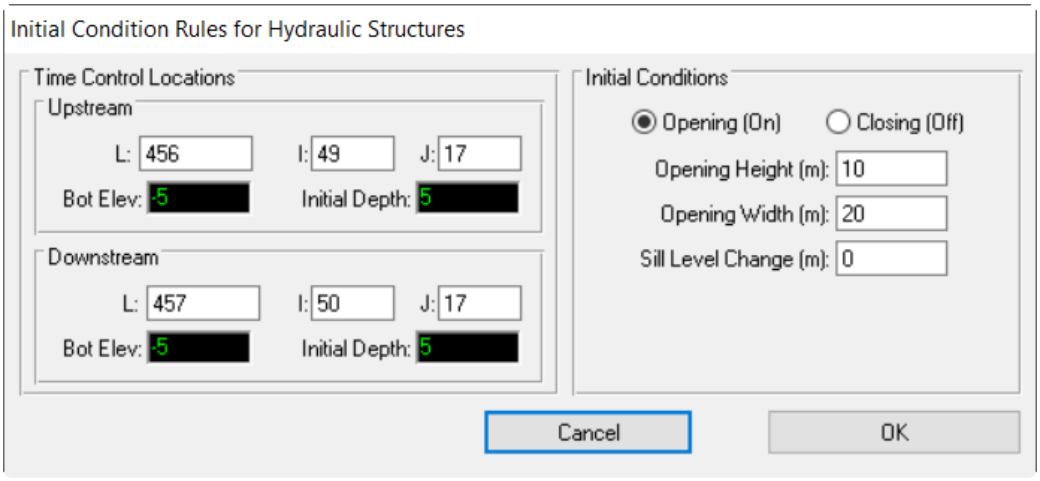


Figure 17. Initial Conditions for gate operation.

4 Rule-based Control for Withdrawal/Return Structures

Withdrawal/return structures have also been enhanced with release of EE8.4. Beside the existing operation of withdrawal/return structures using time-series of flow and constituent rises, a new feature now allows the operation of withdrawal/return structures based on operational rules. This has been implemented in an approach similar to that of the rule-based operations of hydraulic structures. One purpose of this enhancement is to allow the user to simulate rule-based pumps. It should be noted that at this stage this feature only acts on the flow series, and maintains the concentrations as constant. This contrasts with other W/R options that allow the user to set the rise and fall of parameter concentrations.

The interface for the withdrawal/return boundary condition is shown in Figure 18. This form now has a new dropdown menu in the *Current Boundary Group Information* frame, *Time Control*. The user may select from three options: *Controlled using Time-Series*, this was previously the only option; *Controlled using Upstream Elevation,* and *Controlled based on Level Difference.* The time series approach is described in the user manual. The two new approaches require the user to configure control rules for the structure. These rules are set in the *W/R Flow and Concentration Rise/Fall Settings* | *Control Rules*. Here the user may edit existing rule or create new ones as well as set the initial conditions

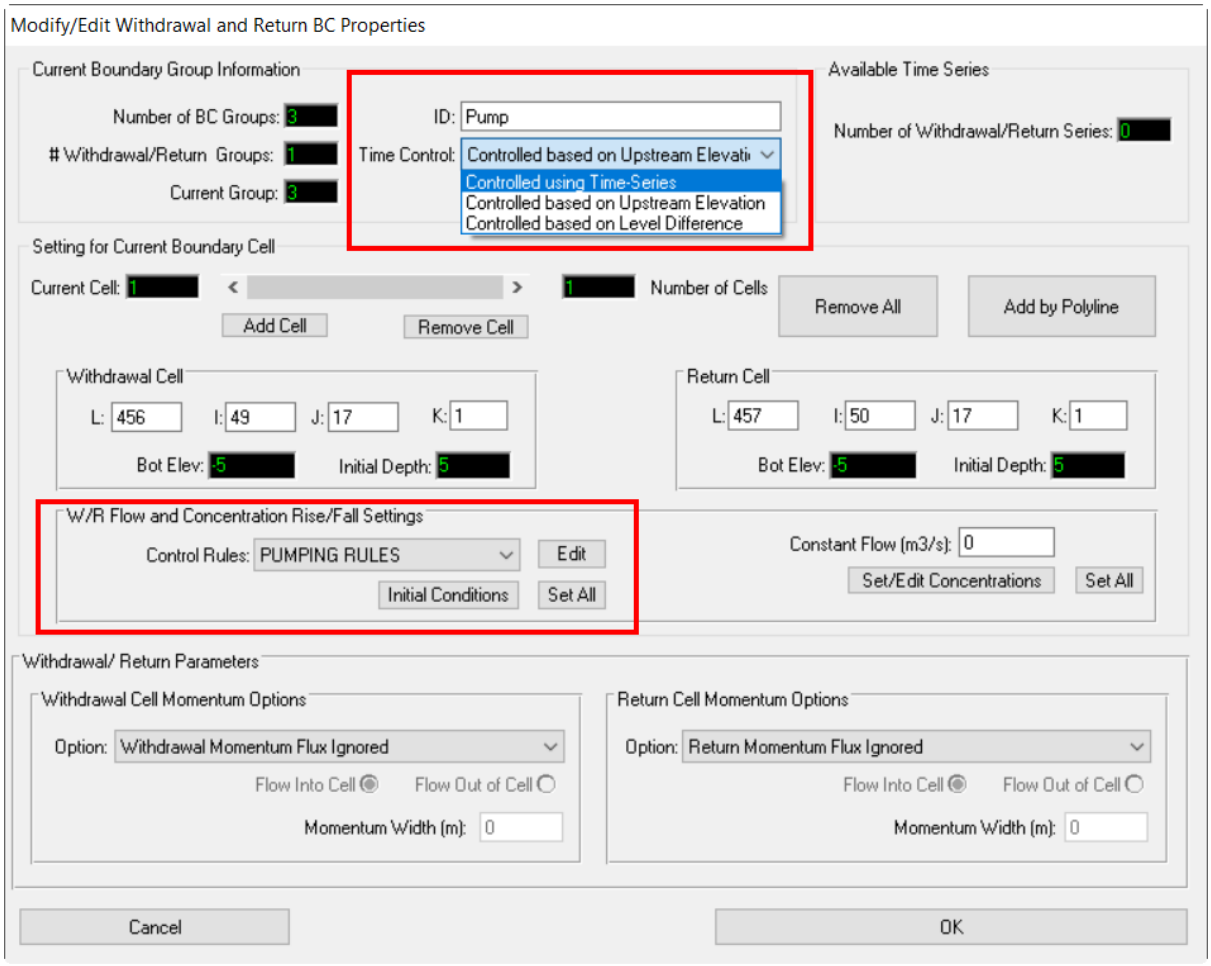


Figure 18. Withdrawal/Return BC: Pumping station settings.

Figure 19 shows the settings for the rule-based withdrawal/return boundary. In this case a pumping station has been configured using the withdrawal/return boundary conditions. The approach is similar to that described for rule-based sluice gate operations: the user configures the number of sets of rules, and uses the up and down arrows to select the *Current Rule* set. Each set can have a number of different triggers, in this case two. The triggers are determined by the control value. This is the upstream elevation, or the difference in upstream and downstream head, depending on which option the user has selected.

EE uses the "state" of the structure to allow the user the high level of control required to simulate realistic scenarios. The logic of the process is that if the state of the structure is "1" then it is on, discharging the *Flow Discharge* set. This will continue from that trigger level until it reaches another trigger level. When a trigger is reached with state of "0" the pump will stop, and flow changes to the new *Flow Discharge.* The prevent instability in the system with sudden turning on and off of large flow rates within one time step, EE allows the user to set a rate of increase and decrease of the flow discharge with the *Rate* variable, in cms/minute.

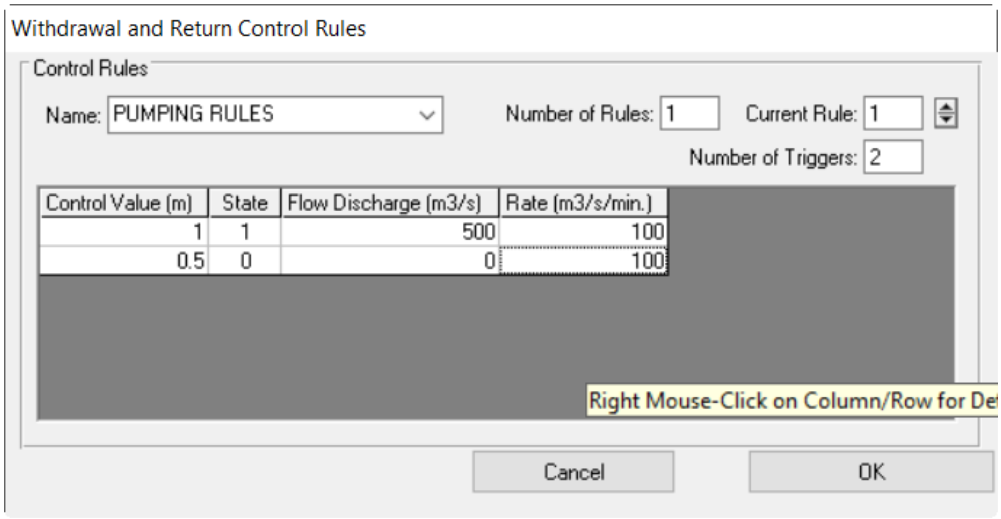


Figure 19. Withdrawal/Return Control Rules GUI.

Figure 20 demonstrates the effect of the operation of the pumping station. In many cases the user will configure a simple system as described here. In this case it would seem the "state" setting is redundant as the pump is simply on or off. However, in some more complex systems there may be two or more pumps at a particular location. These may be configured as lead and lag pumps, with the lead pump turning on at a low water level and lag pump only turning on at a higher level. In these cases the user may want both pumps to stay on even when water level drops below the trigger level of the lead pump. For this reason the "state" setting is required to fully simulate the complexity of the system.

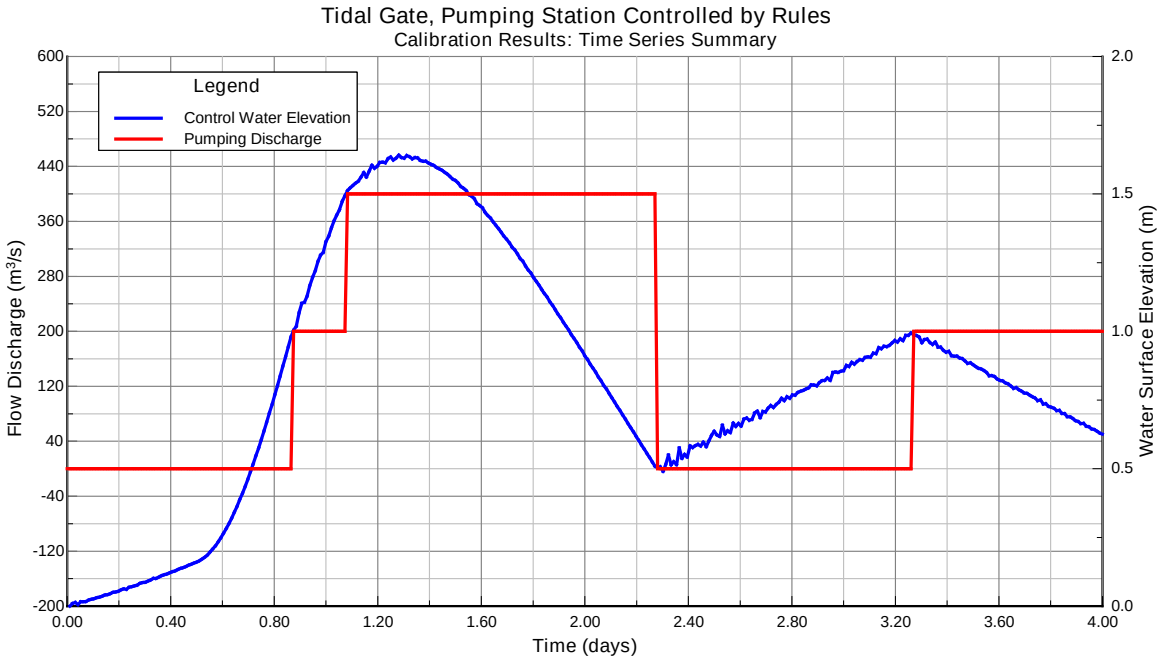


Figure 20. Demonstration of pumping operations.

5 Other Updates

5.1 Spatially Variable AHD and AHO

EEMS8.4 has been enhanced to include user specified spatially variable horizontal eddy viscosity (AHO or Smagorinsky coefficient), and horizontal momentum diffusivity (AHD). This option is useful when the model domain has some regions that have high velocity, and high advectivity, and other that are have low velocity.

There are now three horizontal diffusivity computational options: spatially constant AHO and AHD, spatially variable AHO with constant AHD, and spatially variable AHO and AHD. These options are set under *Hydrodynamics* | *General* | *Modify* as shown in Figure 21 and are described below.

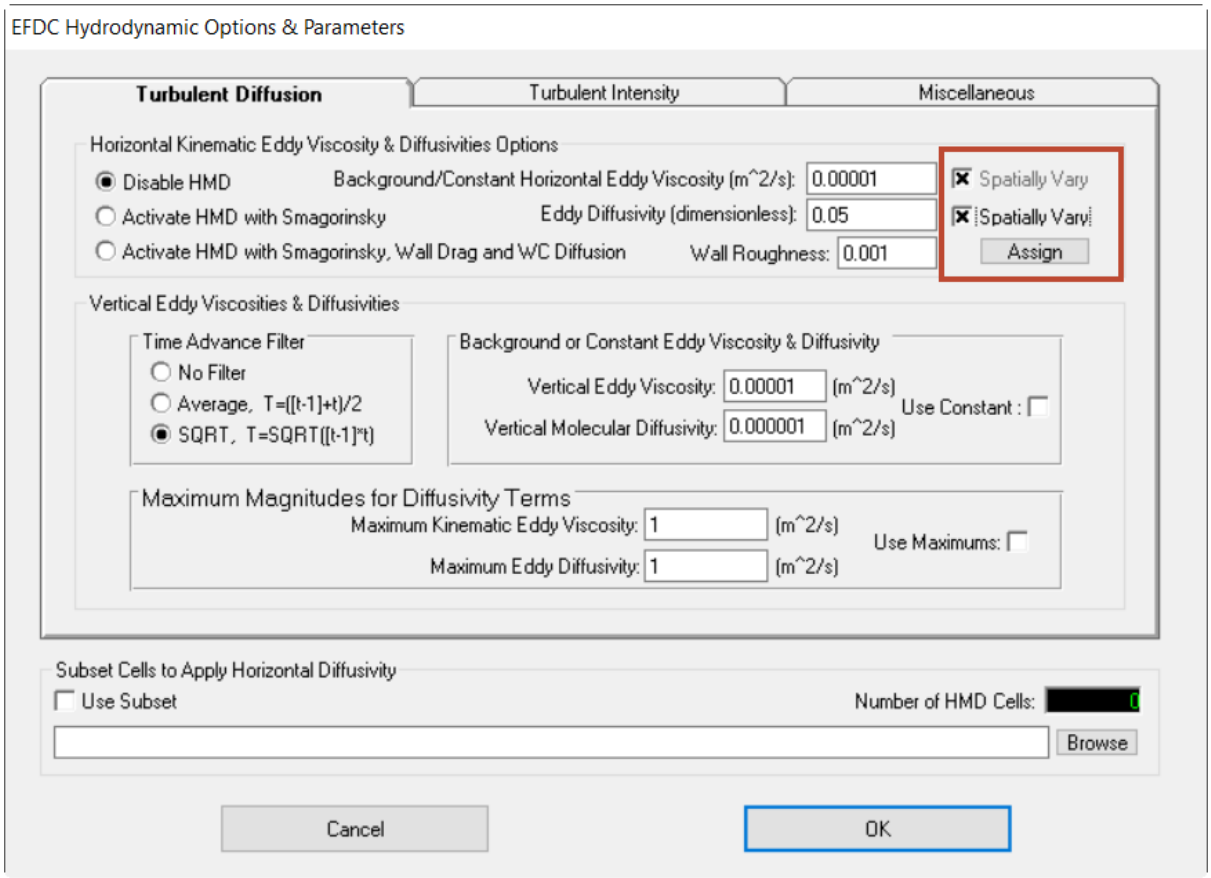


Figure 21. Turbulent diffusion options.

Spatially constant AHO and AHD is the standard EFDC legacy option. In this option neither of the *Spatially Vary* checkboxes are selected.

Spatially variable AHO was added to EE8.3. Previously this option was activated by setting a negative value in the text box. Now there is a *Spatially Vary* checkbox to activate this function. This option calculates the AHO as a function of the cell size:

$$AHO = Factor*DX(L)*DY(L)$$

The “Factor” is the absolute value of the input AHO. In this option AHD is constant.

The new option in EE8.4 is spatially variable AHO and AHD. The user needs to check both of the *Spatially Vary* checkboxes, in which case the *Assign* button is also enabled. Selecting the *Assign* button displays the form shown in Figure 22. EE uses a new file called AHMAP.INP to store a cell by cell list of AHO and AHD. If the AHO in the AHMAP file is a negative number, EFDC+ will use the cell area times the input value to compute the cell's AHO according to the equation above.

Typically, a user wouldn't have data for spatially interpolated AHO and AHD, rather it is expected to use this function to assign constant values to certain regions. The AHO and AHD values can be assigned to the whole model domain using operators or constant values. They can also assign values of AHO and AHD to certain regions using polygon files. Load the polygon file, select the value for AHO or AHD, and select *Apply*. When the user saved the project EE will create an AHMAP.INP file.

To ensure that the correct values have been assigned to the correct regions, the AHO and AHD may also be visualized in *ViewPlan* using the *Fixed Params* viewing option.

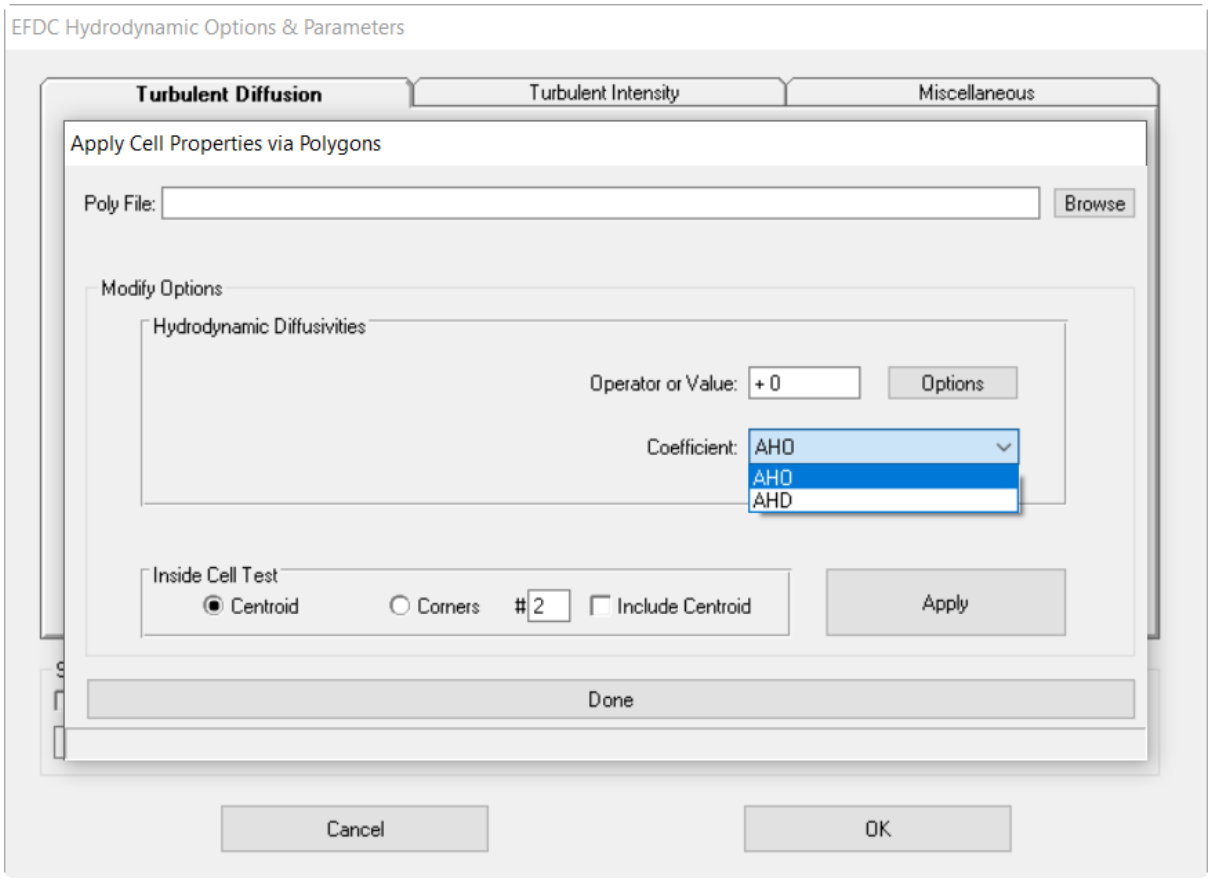


Figure 22. Setting spatially varying AHO and AHD.

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5.2 Momentum Options for the Flow Boundary

EEMS has been enhanced to include a number of new momentum options for the flow boundaries as shown in Figure 23. These are included so that the user has more control over how EFDC handles momentum flux on the cell faces. The options now include:

- Inflow Momentum Flux Ignored (default option)
- Momentum Flux on West U Face
- Momentum Flux on South V Face
- Momentum Flux on East U Face
- Momentum Flux on North V Face

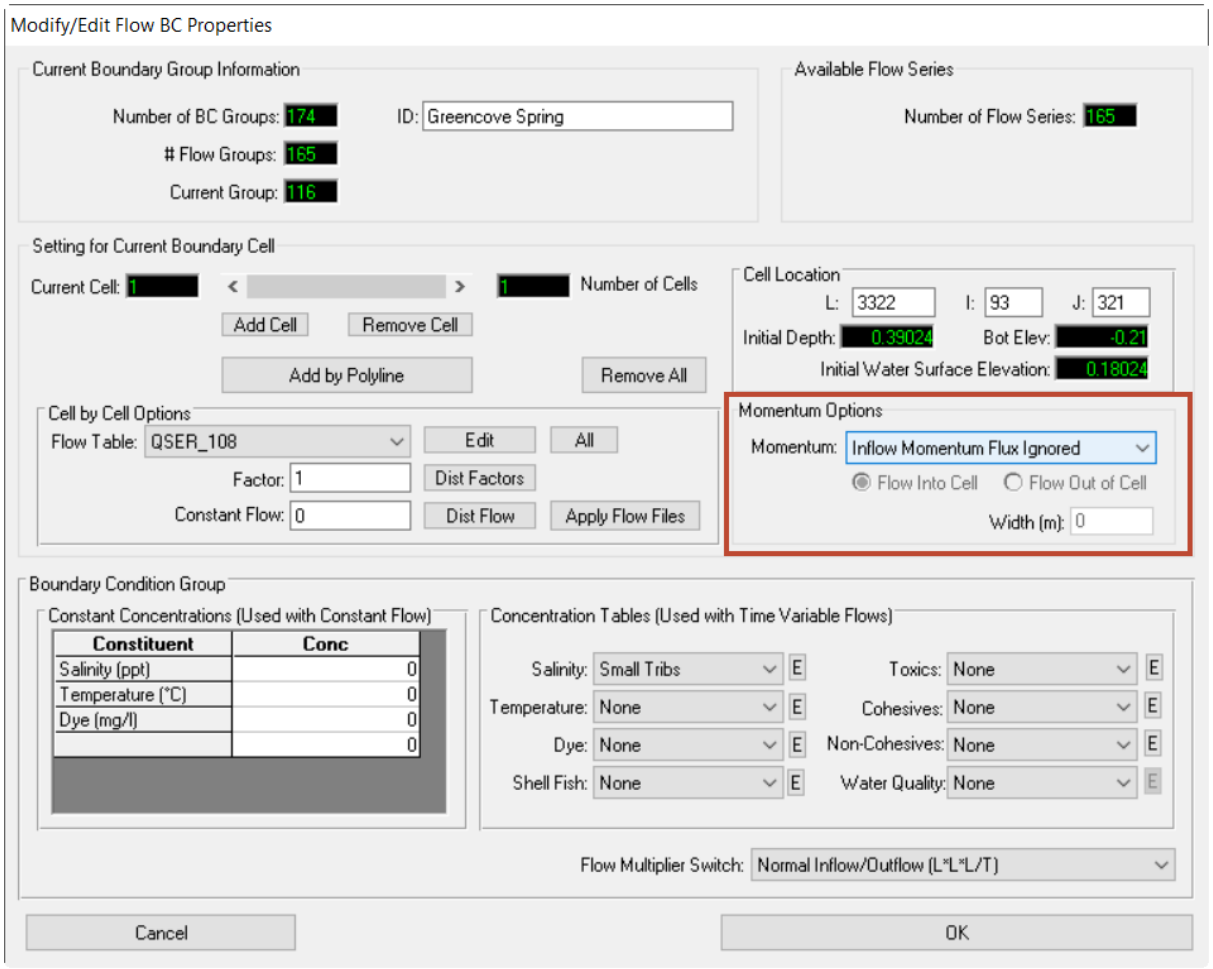


Figure 23. Momentum flux options.

5.3 Elevation Specified Open Boundary Condition

A new boundary condition (BC) option has been which allows the use of an elevation specified flow but allows for a "free tangential". Normally an open BC requires two cells at the border. In some models where there are tight boundaries this can be restrictive. The *Elevation Specified-Free Tangential* option allows EFDC to only use one cell at the border. The types of open boundary condition now supported by EE are:

- Elevation Specified
- Radiation Separation – Zero Tangential
- Radiation Separation – Free Tangential
- Elevation Specified – Free Tangential

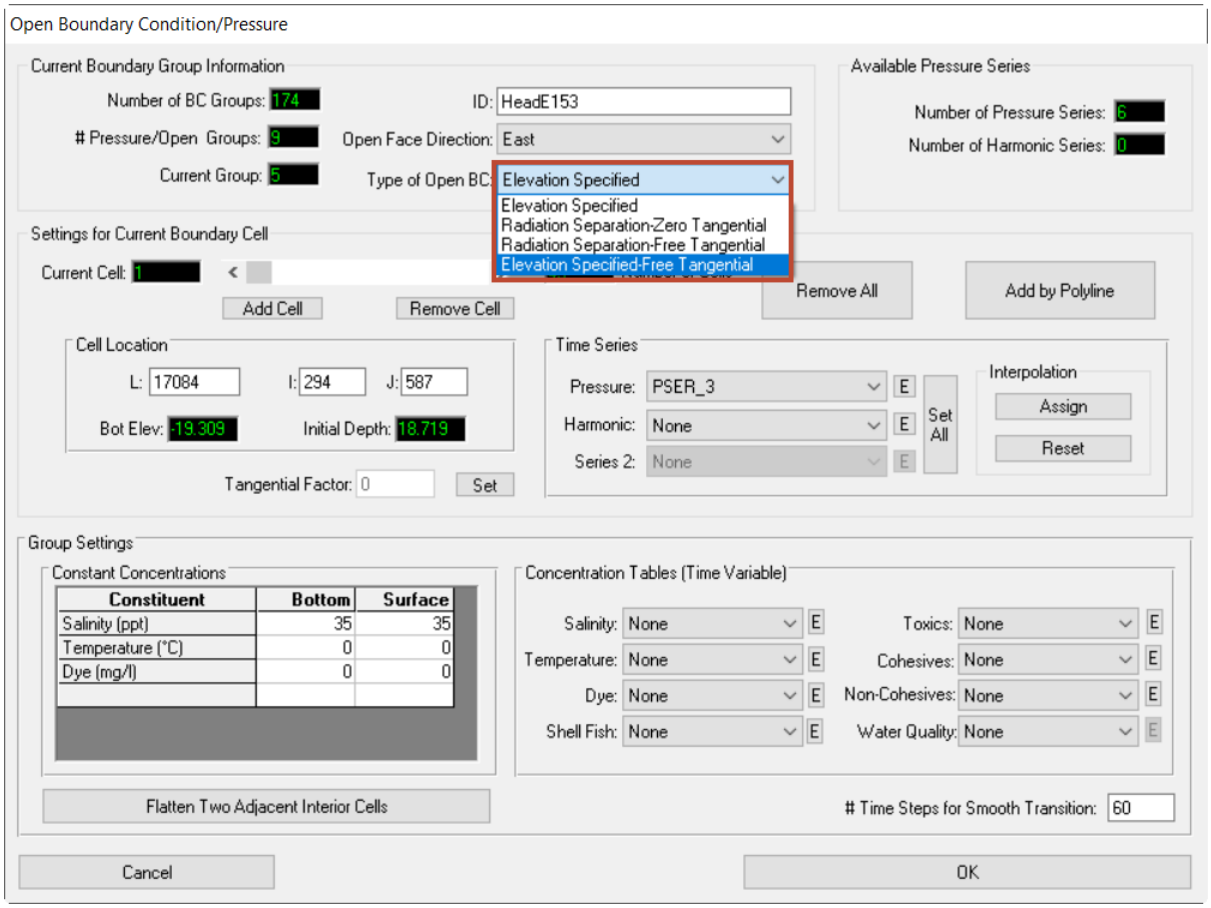


Figure 24. Open BC: Elevation Specified-Free Tangential.

5.4 Whole Channel Rating Curve

EE and EFDC+ have been enhanced to provide the option of setting a boundary that is defined by a single rating curve to represent the river. This feature is useful if the user wants to create stepped flows for a IFIM habitat analysis.

The *Hydraulic Control Structure BC* form is shown in Figure 25. In the *Flow Derived From* dropdown menu the user can select *Upstream Elevation with Whole Channel Rating Curve*. This option acts much like a flow table, or flow time series, which may be split with a flow multiplier across different cells, however in this case it uses the same rating curve for all the cells assigned. Using the scroll bar in the *Setting for Current Boundary Cell* frame allows the user to see the how the multiplier changes.

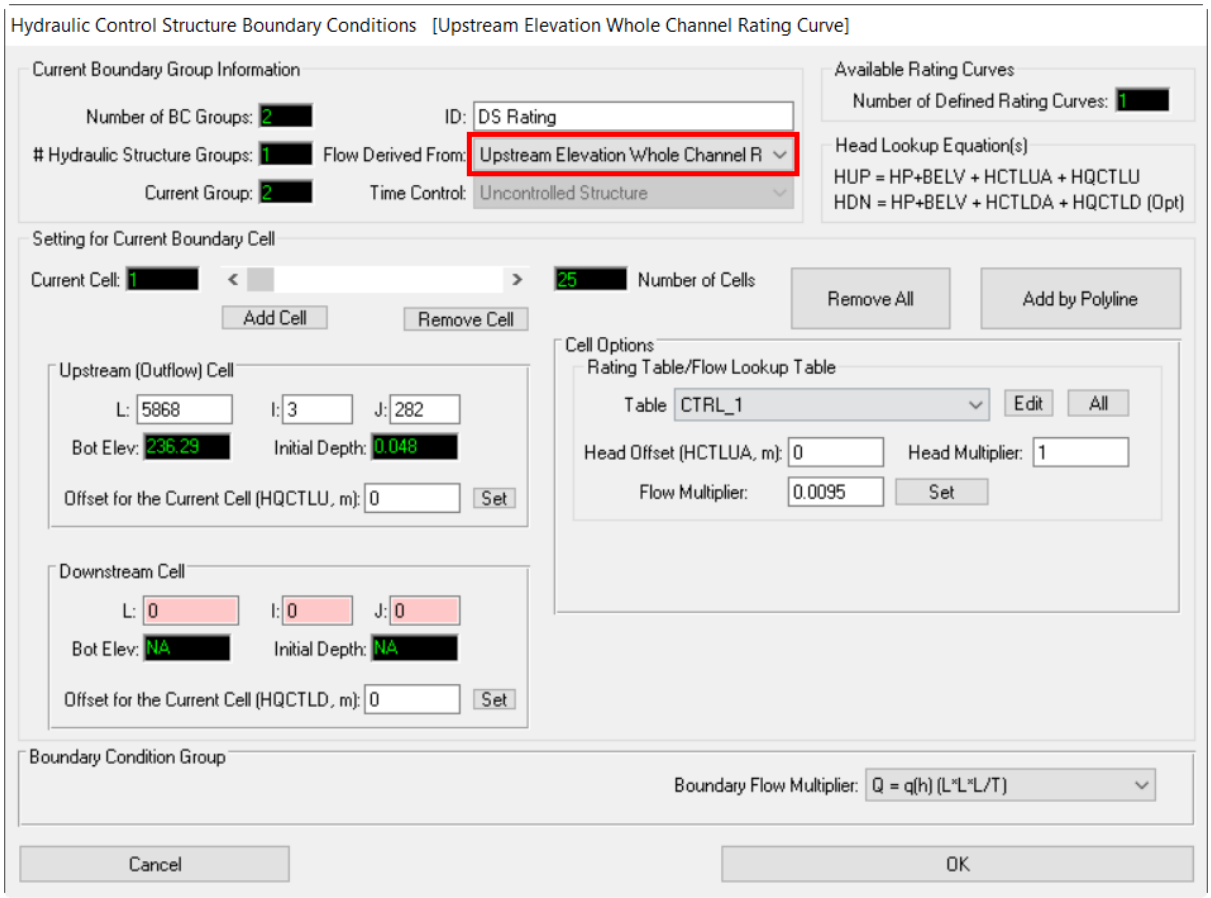


Figure 25. Hydraulic structure: Whole channel rating curve option.

The user also has the option of setting a downstream cell that the flow will return into in the *Downstream Cell* frame. If this is left blank, then EE ignores the downstream end.

When the user selects the *Set* button next to the *Flow Multiplier*, several options are provided for the split approach as shown in Figure 26. The first option is to set the water surface elevation to use when the flow splits are applied. By default this is the maximum value in the rating curve.

The second option is to select one of three approaches for calculating the flow splits. The user may split flow based on DX, on DY, or on DX*DY. In the case shown in Figure 26 the cells are split based on DX.

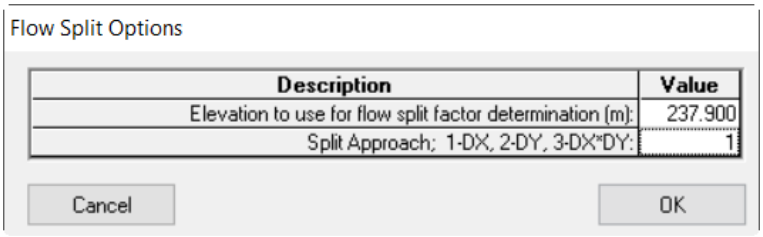


Figure 26. Flow split options.

5.5 Calibration Statistics

When assessing the performance of models in simulating observed conditions at different points in the model domain it is useful to make analyze key statistics. EE8.4 provides a significant enhancement to calibration data to options, allowing them to be processed for sub-periods, and for a wide range of statistics.

Selecting the *Model Analysis* on the main form tab shown in Figure 27, allows the user to select from a range of plots and comparisons: *Time Series Comparisons, Correlation Plots, Vertical Profile Comparisons, High Frequency Time Series, Flux Comparison,* and *Cruise Plot Comparisons*. For each of these options the new statistics that may be reported now include:

- Percent RMS
- Standard Deviation of the Model
- Standard Deviation of the Data,
- R Squared
- Coefficient of Efficiency
- Index of Agreement

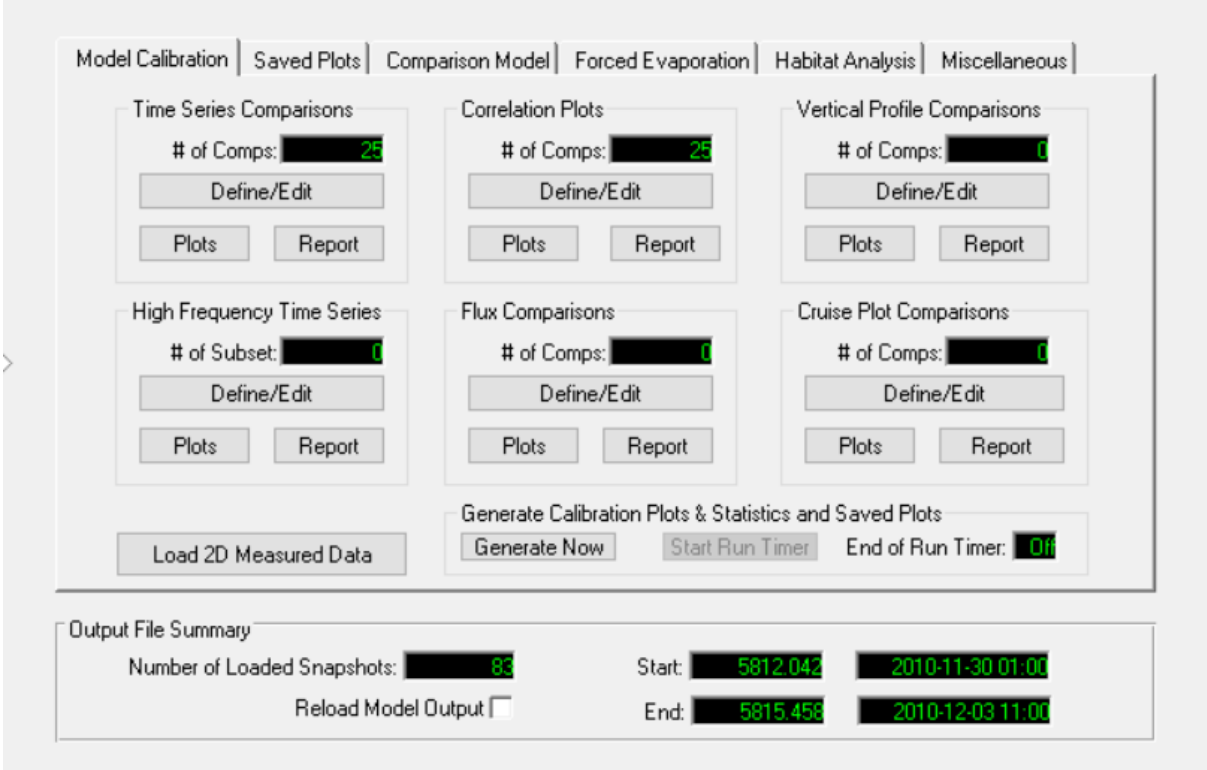


Figure 27. Vertical Profile plots display options.

The difference between the average observation and model values (Mean Error - ME) is taken to show the deviation in terms of the unit of measurement.

RMSE: The root mean squared error (or RMSE) reflects sample standard deviation of differences between simulated and observed values. Similarly, the relative RMSE (RRMSE) is the RMSE normalized by the maximum range of observed values.

The Index of Agreement (IOA) is a standardized measure of the degree of model prediction error, with values ranging from 0 (no agreement) to 1 (perfect match).

The R^2 value is a statistical measure of how close a model's predictions are to a fitted regression line. It is sometimes referred to the coefficient of determination.

To generate the statistics, select the *Report* button (Figure 28) and check the error statistics to be reported. Selecting *Ok* will start the calculations. This can take several minutes for large models. For this reason, the user can select the *Start Run Timer* button which will show the type of plots that EE is generating at that time in the *End of Run Timer*. Once the statistics are displayed, the user can save to a text file or copy to clipboard

The user should select the *Combine Error Statistics* checkbox to plot all the statistics on one form rather than each on a separate form. The user may now also select *HTML Format* so that the statistics are formatted for easy copying into a report or displaying online. To set EE so that if provides automatic summaries of statistics at the end of the model run the user should select *Automatically Generate Reports* checkbox as well as selecting *Generate Plots and Statistic at End of Model Run* in the *Run Model* form before running the model.

Extracting the data from the EFDC output can be time consuming if it is a large model run for a long period of time. To avoid having to extract the results each time the statistics are generated, the user is advised to check the *Use Existing Model Extractions* check box, which will use the extracted data still in computer memory

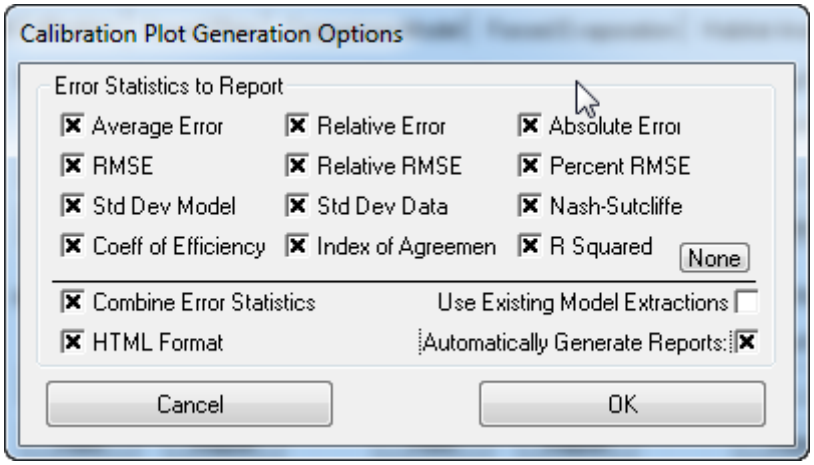


Figure 28. Calibration plot generation options.

The user may apply a similar approach for *Correlation Plots* and *Flux Comparisons.*

Three widely used error measures are used here. “O” denotes observations and “P” denotes model predictions at the corresponding locations and times, the mean of the observed and predicted variables for “N” observations at a single or multiple observation stations is given by:

Mean Observed:

$$\bar{O} = \frac{1}{N}\sum_{i=1}^{N} O_i \tag{B.1}$$

Mean Predicted:

$$\bar{P} = \frac{1}{N}\sum_{i=1}^{N} P_i \tag{B.2}$$

Standard Deviation Observed:

$$SDO = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(O_i - \bar{O}\right)^2} \tag{B.3}$$

Standard Deviation Predicted:

$$SDP = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(P_i - \bar{P}\right)^2} \tag{B.4}$$

Mean Error (ME):

$$ME = \bar{P} - \bar{O} \tag{B.5}$$

Relative Mean Error (RME):

$$RME = \frac{\bar{P} - \bar{O}}{\bar{O}} \times 100\% \tag{B.6}$$

Absolute Mean Error (MAE):

$$MAE = \frac{1}{N}\sum_{i=1}^{N}\left|P_i - O_i\right| \tag{B.7}$$

Root Mean Square Error (RMSE):

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(P_i - O_i\right)^2} \tag{B.8}$$

Relative Root Mean Square Error (RRMSE):

$$RRMSE = \frac{RMSE}{\left(O_{max} - O_{min}\right)} \times 100\% \tag{B.9}$$

(B.10)

Coefficient of Determination (R^2, Square of Correlation Coefficient):

$$R^2 = \frac{\left\{\sum_{i=1}^{N}\left(O_i - \bar{O}\right)\times\left(P_i - \bar{P}\right)\right\}^2}{\left\{\sum_{i=1}^{N}\left(O_i - \bar{O}\right)^2\right\}\times\left\{\sum_{i=1}^{N}\left(P_i - \bar{P}\right)^2\right\}}$$

Nash-Sutcliffe Index of Efficiency (NSI):

$$F^2 = 1 - \frac{\sum_{i=1}^{N}\left(P_i - O_i\right)^2}{\sum_{i=1}^{N}\left(O_i - \bar{O}\right)^2} \qquad \text{(B.11)}$$

Coefficient of Efficiency (COE):

$$E = 1 - \frac{\sum_{i=1}^{N}\left|P_i - O_i\right|}{\sum_{i=1}^{N}\left|O_i - \bar{O}\right|} \qquad \text{(B.12)}$$

Index of Agreement (IOA):

$$D = 1 - \frac{\sum_{i=1}^{N}\left|P_i - O_i\right|}{\sum_{i=1}^{N}\left(\left|P_i - \bar{O}\right| + \left|O_i - \bar{O}\right|\right)} \qquad \text{(B.13)}$$

5.1 Save Plots

EEMS has been enhanced to support automated plot generation of a wide range of graphs, and to facilitate simple and consistent images for reports. A new tab has been added to the *Model Analysis* tab called *Saved Plots* as shown in Figure 29. Plots from *ViewPlan, ViewSlice, View3*, and *2D Longitudinal Profile Plots* can all be configured here.

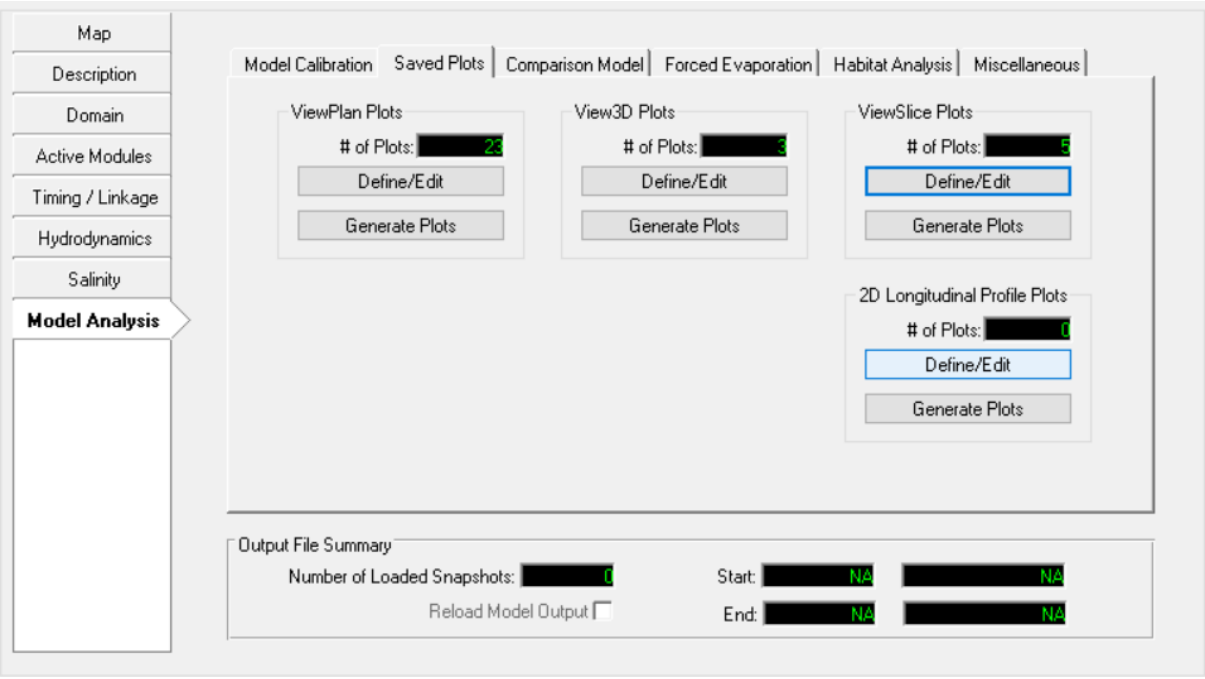


Figure 29. Saved plots options.

5.2 Generic I, J Plots

A feature has been added to *ViewPlan* to allow the user to display any data overlay on the model on a cell by cell basis. This is to support the case for when a user may want to display a particular value on every cell to compare with model output or for reporting purposes. The is especially useful when wanting to view the channel index for habitat analysis. From *ViewPlan* the user should select *Fixed Parameters* in the *Viewing Option* sidebar. When this is selected EE prompts for an input file. This file should be of the format: I, J, value. This file may be comma or tab delimited. The options button allows the user to enter the parameter name and units. If this is left blank then EE will prompt the user for this information. EE retains the link to this data when the model is closed or saved for convenience when reopening the *Generic I J value*. It may be changed by browsing to another file.

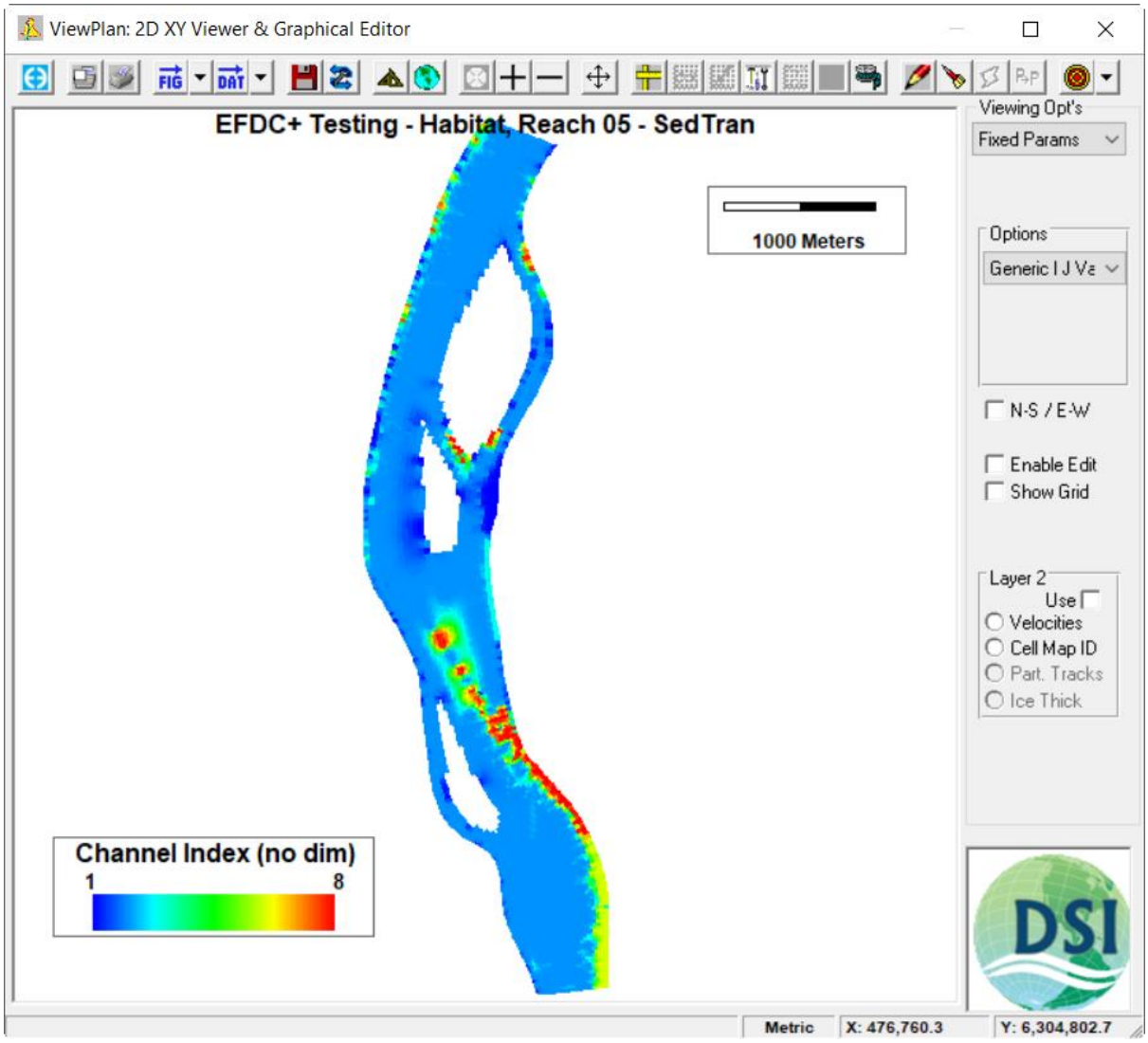


Figure 30. Generic I, J option displaying channel index.

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Bovee, K.D 1982. A Guide to Stream Habitat Analysis using the Instream Flow Incremental Methodology. Instream Flow Inf. Pap. 12. U.S. Fish Wildlife Servo FWS/OBS-82/26. 248 pp

Bovee, K.D., Lamb, B.L., Bartholow, J.M., Stalnaker, C.B., Taylor, J., Henriksen, J., 1998, Stream habitat analysis using the instream flow incremental methodology: U.S. Geological Survey, Biological Resources Division Information and Technology Report USGS/BRD-1998-0004. 131p.

Milhous, R.T. and T.J. Waddle. 2012. Physical Habitat Simulation (PHABSIM) Software for Windows. Fort Collins, CO: USGS Fort Collins Science Center